Quant Interview Drill Guide

Casella & Berger Chapters 7–9 + Linear Regression

High-Impact Outline: What to Drill Until Automatic

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1 Point Estimation (C&B Chapter 7) — The Heavy Lifter

Spend 50% of your time here. This is the bread and butter of quant stats interviews.

1.1 Method of Moments (MOM)

Let $X_1, \ldots, X_n$ be iid from a distribution with k unknown parameters $\theta_1, \ldots, \theta_k$. The MOM equates sample moments to population moments. Define sample moments $m_j = \frac{1}{n}\sum_{i=1}^n X_i^j$ and population moments $\mu'_j = \mathbb{E}[X^j]$ for $j = 1, \ldots, k$. Set $m_j = \mu'_j(\theta_1, \ldots, \theta_k)$ and solve the resulting $k \times k$ system.

For one-parameter families this is especially simple. For Uniform$(0, \theta)$: $\mathbb{E}[X] = \theta/2$, so $\bar{X} = \theta/2$, giving $\hat{\theta}_{\text{MOM}} = 2\bar{X}$.

1.2 Maximum Likelihood Estimation (MLE)

Given iid data $X_1, \ldots, X_n$ with pdf/pmf $f(x|\theta)$, the MLE maximizes the likelihood $L(\theta|\mathbf{x}) = \prod_{i=1}^n f(x_i|\theta)$. The standard approach:

1. Take the log-likelihood $\ell(\theta) = \sum_{i=1}^n \log f(x_i|\theta)$.
2. Differentiate and set to zero: $\ell'(\theta) = 0$.
3. Solve for $\hat{\theta}$ and verify it is a maximum.

Warning: Check the Support Before Differentiating

If the support of $f(x|\theta)$ depends on θ, the log-likelihood may not have an interior critical point. You must maximize L directly—often the MLE is at a boundary.

The indicator function trap. This is a favorite interview question. Let $X_1, \ldots, X_n \sim$ iid Uniform$(0, \theta)$. The pdf is $f(x|\theta) = \theta^{-1}\mathbf{1}_{(0,\theta)}(x)$, so the likelihood is

$$L(\theta|\mathbf{x}) = \prod_{i=1}^n \frac{1}{\theta}\mathbf{1}_{(0,\theta)}(x_i) = \frac{1}{\theta^n}\prod_{i=1}^n \mathbf{1}_{(0,\theta)}(x_i).$$

The product of indicators is 1 iff all $x_i \in (0, \theta)$, i.e. iff $\theta \geq x_{(n)}$. So

$$L(\theta|\mathbf{x}) = \begin{cases} \theta^{-n} & \theta \geq x_{(n)}, \\ 0 & \theta < x_{(n)}. \end{cases}$$

For $\theta \geq x_{(n)}$, L is strictly decreasing in θ. The maximum is at the boundary:

$$\hat{\theta}_{\text{MLE}} = X_{(n)} = \max(X_1, \ldots, X_n).$$

Calculus gives $dL/d\theta = -n\theta^{-(n+1)} < 0$ for all $\theta > x_{(n)}$—it is never zero. The maximum is at a boundary, not an interior critical point.

Invariance. If $\hat{\theta}$ is the MLE of θ, then $\widehat{\tau(\theta)} = \tau(\hat{\theta})$ for any function τ.

1.3 Bias, Variance, and Mean Squared Error (MSE)

The mean squared error (MSE) of an estimator W for θ decomposes as

$$\mathrm{MSE}(W) = \mathbb{E}\Big[(W-\theta)^2\Big] = \mathrm{Var}(W) + (\mathrm{Bias})^2, \qquad \mathrm{Bias} = \mathbb{E}[W] - \theta.$$

Derivation. Write $W - \theta = (W - \mathbb{E}[W]) + (\mathbb{E}[W] - \theta)$. Squaring and taking expectations, the cross term vanishes because $\mathbb{E}[W - \mathbb{E}[W]] = 0$:

$$\mathbb{E}\Big[(W-\theta)^2\Big] = \mathbb{E}\Big[(W-\mathbb{E}[W])^2\Big] + (\mathbb{E}[W]-\theta)^2 = \mathrm{Var}(W) + (\mathrm{Bias})^2.$$

If W is unbiased then $\mathrm{MSE} = \mathrm{Var}(W)$. But a biased estimator can have smaller MSE if the variance reduction outweighs the bias penalty.

Warning: Biased Can Beat Unbiased

The MLE $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ has smaller MSE than the unbiased $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ for normal data. When asked "which estimator is better?" always ask: better in what sense? Unbiasedness is not the only criterion.

1.4 Sufficiency and the Factorization Theorem

A sufficient statistic $T(\mathbf{X})$ captures all the information in the sample about θ.

Factorization Theorem (Fisher–Neyman). $T(\mathbf{X})$ is sufficient for θ iff the joint pdf/pmf factors as

$$f(\mathbf{x}|\theta) = g(T(\mathbf{x}),\, \theta) \cdot h(\mathbf{x}),$$

where g depends on the data only through $T(\mathbf{x})$, and h depends only on the data.

How to use it: write the joint density, identify which terms involve θ, and read off which functions of $x_1, \ldots, x_n$ appear in those terms. For example, for $\mathrm{Exp}(\lambda)$:

$$f(\mathbf{x}|\lambda) = \lambda^n \exp(-\lambda \textstyle\sum x_i) \cdot 1,$$

so $T = \sum X_i$ is sufficient. For $\mathrm{Uniform}(0, \theta)$:

$$f(\mathbf{x}|\theta) = \underbrace{\frac{1}{\theta^n}\mathbf{1}_{(0,\theta)}(x_{(n)})}_{g(x_{(n)},\,\theta)} \cdot \underbrace{1}_{h(\mathbf{x})},$$

so $T = X_{(n)}$ is sufficient (the max carries all the information about θ).

Exponential family shortcut. If $f(x|\boldsymbol{\theta}) = h(x)\, c(\boldsymbol{\theta}) \exp\big(\sum_{j=1}^k w_j(\boldsymbol{\theta})\, t_j(x)\big)$, then $T = \big(\sum t_1(X_i), \ldots, \sum t_k(X_i)\big)$ is complete sufficient.

1.5 Rao–Blackwell Theorem

If W is any unbiased estimator of $\tau(\theta)$ and T is sufficient for θ, then $\phi(T) = \mathbb{E}[W \mid T]$ is:

1. still unbiased: $\mathbb{E}[\phi(T)] = \mathbb{E}[\mathbb{E}[W|T]] = \mathbb{E}[W] = \tau(\theta)$;
2. at least as good: $\mathrm{Var}(\phi(T)) \le \mathrm{Var}(W)$.

The variance reduction follows from the law of total variance:

$$\mathrm{Var}(W) = \mathrm{Var}(\mathbb{E}[W|T]) + \mathbb{E}[\mathrm{Var}(W|T)] = \mathrm{Var}(\phi(T)) + \underbrace{\mathbb{E}[\mathrm{Var}(W|T)]}_{\geq 0}.$$

So $\mathrm{Var}(W) \geq \mathrm{Var}(\phi(T))$, with equality iff W is already a function of T.

If T is also **complete**, then $\phi(T)$ is the unique UMVUE (uniformly minimum variance unbiased estimator) of $\tau(\theta)$.

1.6 Cramér–Rao Lower Bound (CRLB)

Under regularity conditions (support does not depend on θ, interchange of differentiation and integration is valid), any unbiased estimator W of $\tau(\theta)$ satisfies

$$\mathrm{Var}(W) \geq \frac{[\tau'(\theta)]^2}{n\, I(\theta)},$$

where $I(\theta) = \mathbb{E}\Big[\big(\frac{\partial}{\partial\theta} \log f(X|\theta)\big)^2\Big]$ is the Fisher information for one observation.

Alternative form of Fisher information. For exponential families, the second-derivative form is often easier to compute:

$$I(\theta) = -\mathbb{E}\left[\frac{\partial^2}{\partial\theta^2} \log f(X|\theta)\right].$$

To see why these are equal, start from $\int f(x|\theta)\, dx = 1$. Differentiate both sides with respect to θ:

$$\int \frac{\partial}{\partial\theta} f(x|\theta)\, dx = 0 \quad \Longrightarrow \quad \mathbb{E}\left[\frac{\partial}{\partial\theta} \log f(X|\theta)\right] = 0,$$

using the identity $\frac{\partial}{\partial\theta} \log f = \frac{1}{f}\frac{\partial f}{\partial\theta}$. Differentiating again:

$$0 = \int \frac{\partial^2 f}{\partial\theta^2}\, dx = \int \left[\frac{\partial^2 \log f}{\partial\theta^2} + \left(\frac{\partial \log f}{\partial\theta}\right)^2\right] f\, dx = \mathbb{E}\left[\frac{\partial^2 \log f}{\partial\theta^2}\right] + I(\theta).$$

Rearranging gives $I(\theta) = -\mathbb{E}\Big[\frac{\partial^2}{\partial\theta^2} \log f\Big]$.

Warning: When the CRLB Does Not Apply

For Uniform$(0, \theta)$, the support depends on θ, so the regularity conditions fail. The UMVUE $\frac{n+1}{n} X_{(n)}$ has variance $\theta^2/(n(n+2)) = O(1/n^2)$, which is faster than any $O(1/n)$ bound the CRLB could give. Don't reflexively apply the CRLB without checking regularity.

1.7 Summary: MLE vs. MOM for Common Distributions

Distribution	Parameter(s)	MOM	MLE
$N(\mu, \sigma^2)$	μ, σ^2	$\bar{X},\ \frac{1}{n}\sum(X_i - \bar{X})^2$	$\bar{X},\ \frac{1}{n}\sum(X_i - \bar{X})^2$
Exp(λ)	λ (rate)	$1/\bar{X}$	$1/\bar{X}$
Poisson(λ)	λ	$\bar{X}$	$\bar{X}$
Uniform$(0, \theta)$	θ	$2\bar{X}$	$X_{(n)}$
Bernoulli(p)	p	$\bar{X}$	$\bar{X}$
Gamma(α, β)	α, β	See problems	No closed form
Beta(α, β)	α, β	See problems	No closed form

2 Hypothesis Testing (C&B Chapter 8) — The Logic

Interviews focus on the setup of the test, not the philosophical implications.

2.1 Test Setup

Every hypothesis test starts with two competing statements about a parameter θ: the **null hypothesis** $H_0 : \theta \in \Theta_0$ (the status quo) and the **alternative** $H_1 : \theta \in \Theta_0^c$ (what we want evidence for). A hypothesis is **simple** if it pins down one distribution ($\theta = \theta_0$), **composite** if it specifies several ($\theta > \theta_0$).

A **test** partitions the sample space into a rejection region R (reject H_0) and its complement. We specify it through a test statistic $W(\mathbf{X})$ and a cutoff.

Type I Error, Type II Error, Size, and Power

	Accept H_0	**Reject H_0**
H_0 true	Correct	**Type I Error**
H_1 true	**Type II Error**	Correct

The **power function** is $\beta(\theta) = \mathbb{P}(\mathbf{X} \in R \mid \theta)$.
A test has **size** α if $\sup_{\theta \in \Theta_0} \beta(\theta) = \alpha$.
A test has **level** α if $\sup_{\theta \in \Theta_0} \beta(\theta) \leq \alpha$.

Warning: Size vs. Level

Size α means the supremum *equals* α. Level α means it is *at most* α. Every size α test is level α, but not conversely.

Warning: One-Sided vs. Two-Sided: Get the Direction Right

- $H_1 : \theta > \theta_0 \Rightarrow$ reject for *large* $\bar{X}$ (right tail).
- $H_1 : \theta < \theta_0 \Rightarrow$ reject for *small* $\bar{X}$ (left tail).
- $H_1 : \theta \neq \theta_0 \Rightarrow$ reject for $|\bar{X} - \theta_0|$ large (both tails).

Mixing up the rejection direction is one of the most common interview mistakes.

2.2 Likelihood Ratio Tests (LRT)

The LRT is the most general-purpose method for constructing tests. For $H_0 : \theta \in \Theta_0$ versus $H_1 : \theta \in \Theta_0^c$, define

$$\lambda(\mathbf{x}) = \frac{\sup_{\theta \in \Theta_0} L(\theta \mid \mathbf{x})}{\sup_{\theta \in \Theta} L(\theta \mid \mathbf{x})}.$$

Since the numerator is a restricted maximum, $0 \leq \lambda(\mathbf{x}) \leq 1$. Small λ means the data are much more likely under some value in Θ_0^c than under any value in Θ_0, so we reject H_0 when $\lambda(\mathbf{x}) \leq c$.

Recipe. (1) Find the unrestricted MLE $\hat{\theta}$ over Θ (denominator). (2) Find the restricted MLE $\hat{\theta}_0$ over Θ_0 (numerator). (3) Form $\lambda = L(\hat{\theta}_0)/L(\hat{\theta})$. (4) Simplify $\lambda \le c$ into a condition on a familiar statistic. (5) Choose c for size α.

If $T(\mathbf{X})$ is sufficient for θ, then $\lambda(\mathbf{x})$ depends on $\mathbf{x}$ only through $T(\mathbf{x})$.

Deriving the z-test from the LRT. Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\theta, \sigma^2)$ with σ^2 known. Test $H_0\colon \theta = \theta_0$ vs. $H_1\colon \theta \neq \theta_0$.

The normal likelihood for a single θ value is

$$L(\theta|\mathbf{x}) = (2\pi\sigma^2)^{-n/2} \exp\Big[-\frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i-\theta)^2\Big].$$

The unrestricted MLE is $\hat{\theta} = \bar{x}$ (maximizes L over all $\theta \in \mathbb{R}$). The restricted MLE under H_0 is $\hat{\theta}_0 = \theta_0$ (the only value in Θ_0). The LRT statistic is

$$\lambda(\mathbf{x}) = \frac{L(\theta_0|\mathbf{x})}{L(\bar{x}|\mathbf{x})} = \frac{\exp\big[-\frac{1}{2\sigma^2}\sum(x_i-\theta_0)^2\big]}{\exp\big[-\frac{1}{2\sigma^2}\sum(x_i-\bar{x})^2\big]}.$$

The $(2\pi\sigma^2)^{-n/2}$ cancels. Now use the identity $\sum(x_i-\theta_0)^2 = \sum(x_i-\bar{x})^2 + n(\bar{x}-\theta_0)^2$ (completing the square). Substituting:

$$\lambda(\mathbf{x}) = \exp\Big[-\frac{1}{2\sigma^2}\Big(\sum(x_i-\bar{x})^2 + n(\bar{x}-\theta_0)^2 - \sum(x_i-\bar{x})^2\Big)\Big] = \exp\Big[-\frac{n(\bar{x}-\theta_0)^2}{2\sigma^2}\Big].$$

Now convert the rejection rule. Rejecting when $\lambda \le c$ means

$$\exp\Big[-\frac{n(\bar{x}-\theta_0)^2}{2\sigma^2}\Big] \le c \quad\Longleftrightarrow\quad \frac{n(\bar{x}-\theta_0)^2}{\sigma^2} \ge -2\log c \quad\Longleftrightarrow\quad \left|\frac{\bar{x}-\theta_0}{\sigma/\sqrt{n}}\right| \ge \sqrt{-2\log c}.$$

Define $Z = (\bar{X}-\theta_0)/(\sigma/\sqrt{n})$. Under H_0, $Z \sim \mathrm{n}(0,1)$, so choosing c for size α gives the familiar rule: reject when $|Z| \ge z_{\alpha/2}$.

Warning: The LRT Does Not Always Give a "Nice" Test

The LRT may not have a closed-form rejection region, may not be the most powerful test, and can require substantial algebra when nuisance parameters are present.

2.3 The Neyman–Pearson (NP) Lemma

For testing $H_0\colon \theta = \theta_0$ versus $H_1\colon \theta = \theta_1$ (both simple), the NP Lemma says: the test that rejects when

$$\frac{f(\mathbf{x} \mid \theta_1)}{f(\mathbf{x} \mid \theta_0)} > k$$

(with k chosen so the size is α) is the **uniformly most powerful (UMP)** level α test.

Proof. Let ϕ be the NP test and ϕ' any other level α test. By the NP rejection rule, for every $\mathbf{x}$:

- If $\phi(\mathbf{x}) = 1$ (reject), then $f(\mathbf{x}|\theta_1) \ge k\, f(\mathbf{x}|\theta_0)$.
- If $\phi(\mathbf{x}) = 0$ (accept), then $f(\mathbf{x}|\theta_1) \le k\, f(\mathbf{x}|\theta_0)$.

In both cases, $(\phi - \phi')(f(\mathbf{x}|\theta_1) - k\, f(\mathbf{x}|\theta_0)) \geq 0$. Integrating:

$$0 \leq \int [\phi - \phi'][f(\mathbf{x}|\theta_1) - k\, f(\mathbf{x}|\theta_0)]\, d\mathbf{x} = [\beta(\theta_1) - \beta'(\theta_1)] - k\underbrace{[\alpha - \beta'(\theta_0)]}_{\geq 0}.$$

Since $\beta(\theta_0) = \alpha$ and $\beta'(\theta_0) \leq \alpha$, the second bracket is ≥ 0, so $\beta(\theta_1) \geq \beta'(\theta_1)$. No other level α test has higher power at θ_1. □

2.4 Power Functions

For the one-sided z-test ($H_0\colon \theta \leq \theta_0$, $H_1\colon \theta > \theta_0$, σ^2 known), the power function is

$$\beta(\theta) = \mathbb{P}\left(\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} > z_\alpha\right) = 1 - \Phi\left(z_\alpha + \frac{\theta_0 - \theta}{\sigma/\sqrt{n}}\right).$$

Derivation. Under θ, $\bar{X} \sim \mathrm{n}(\theta, \sigma^2/n)$. The test statistic is centered at θ_0, not θ, so it is not standard normal. Subtract and add θ:

$$\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} = \frac{(\bar{X} - \theta) + (\theta - \theta_0)}{\sigma/\sqrt{n}} = \underbrace{\frac{\bar{X} - \theta}{\sigma/\sqrt{n}}}_{\sim \mathrm{n}(0,1)} + \frac{\theta - \theta_0}{\sigma/\sqrt{n}}.$$

So the test statistic is $\mathrm{n}(\delta, 1)$ where $\delta = (\theta - \theta_0)/(\sigma/\sqrt{n})$ is the "signal-to-noise" shift. Writing Z for a standard normal:

$$\mathbb{P}\left(\frac{\bar{X} - \theta_0}{\sigma/\sqrt{n}} > z_\alpha\right) = \mathbb{P}(Z + \delta > z_\alpha) = \mathbb{P}(Z > z_\alpha - \delta) = 1 - \Phi(z_\alpha - \delta).$$

Key properties: $\beta(\theta)$ is increasing in θ; $\beta(\theta_0) = \alpha$; power increases with n.

A test is **unbiased** if $\beta(\theta') \geq \beta(\theta'')$ for every $\theta' \in \Theta_0^c$ and $\theta'' \in \Theta_0$ (power under the alternative is never less than under the null).

2.5 Monotone Likelihood Ratio and Karlin–Rubin

A family $\{g(t|\theta)\}$ for sufficient statistic T has **monotone likelihood ratio (MLR)** if for every $\theta_2 > \theta_1$, the ratio $g(t|\theta_2)/g(t|\theta_1)$ is nondecreasing in t. Most "nice" families have MLR: Normal, Binomial, Poisson, Exponential, and any one-parameter exponential family with $w(\theta)$ increasing.

Karlin–Rubin Theorem. If T is sufficient with MLR, then for $H_0\colon \theta \leq \theta_0$ vs. $H_1\colon \theta > \theta_0$, the test that rejects when $T > t_0$ (with $P_{\theta_0}(T > t_0) = \alpha$) is UMP level α.

Why it works. Fix any $\theta_1 > \theta_0$. By Neyman–Pearson, the best test of θ_0 vs. θ_1 rejects for large $g(t|\theta_1)/g(t|\theta_0)$. By MLR, large ratio $\Leftrightarrow$ large t. The cutoff t_0 depends only on α and θ_0, not on which θ_1 we pick. So the same test is most powerful against every $\theta_1 > \theta_0$—it is UMP.

Warning: Two-Sided Tests: No UMP

For $H_1\colon \theta \neq \theta_0$, UMP level α tests typically do not exist. The NP tests for $\theta_1 > \theta_0$ and $\theta_1 < \theta_0$ point in opposite directions. The standard approach is UMP *unbiased* (UMPU) tests.

2.6 p-Values

A p-value $p(\mathbf{x})$ satisfies $P_\theta(p(\mathbf{X}) \le \alpha) \le \alpha$ for every $\theta \in \Theta_0$. If large values of $W(\mathbf{X})$ give evidence against H_0, the standard construction is

$$p(\mathbf{x}) = \sup_{\theta \in \Theta_0} P_\theta\big(W(\mathbf{X}) \ge W(\mathbf{x})\big).$$

2.7 Sample Size Determination

For a one-sided z-test with size α and power $1 - \beta^*$ at a specific alternative $\theta_1 > \theta_0$:

$$n = \left\lceil \frac{(z_\alpha + z_{\beta^*})^2\, \sigma^2}{(\theta_1 - \theta_0)^2} \right\rceil.$$

Derivation. We need $\beta(\theta_1) = 1 - \beta^*$. Using the power function derived above with $\delta_1 = (\theta_1 - \theta_0)/(\sigma/\sqrt{n})$:

$$1 - \Phi(z_\alpha - \delta_1) = 1 - \beta^*.$$

Applying Φ to both sides: $z_\alpha - \delta_1 = -z_{\beta^*}$ (since $\Phi(-z_{\beta^*}) = \beta^*$). Rearranging:

$$\delta_1 = z_\alpha + z_{\beta^*} \quad \Longrightarrow \quad \frac{(\theta_1 - \theta_0)\sqrt{n}}{\sigma} = z_\alpha + z_{\beta^*} \quad \Longrightarrow \quad n = \frac{(z_\alpha + z_{\beta^*})^2\, \sigma^2}{(\theta_1 - \theta_0)^2}.$$

2.8 Summary: Which Tool When?

Situation	**Method**	**Result**
Simple vs. simple	Neyman–Pearson	UMP test
One-sided, MLR family	Karlin–Rubin	UMP test
Two-sided, one parameter	NP / UMPU theory	UMP unbiased test
General composite	LRT	Good (often asymptotically optimal)
Unknown nuisance parameters	LRT	Often reduces to t- or F-test

3 Interval Estimation (C&B Chapter 9) — The Mechanics

The quickest chapter. If you can do Ch. 8, you can do Ch. 9.

A confidence interval (CI) $[L(\mathbf{X}), U(\mathbf{X})]$ is a random interval—the data are random, the parameter is fixed. The confidence coefficient is $\inf_\theta \mathbb{P}(\theta \in [L(\mathbf{X}), U(\mathbf{X})]) \geq 1 - \alpha$.

Warning: The Interval Is Random, Not the Parameter

"95% CI" means: if we repeated the experiment many times, 95% of the resulting intervals would contain the true θ. Saying "θ has a 95% probability of being in the interval" is a Bayesian credible set statement, not a frequentist CI statement. Common interview trap.

3.1 Pivotal Quantities

A random variable $Q(\mathbf{X}, \theta)$ is a **pivot** if its distribution does not depend on any unknown parameters. If we can find constants a, b (depending only on α) with $\mathbb{P}(a \leq Q \leq b) = 1 - \alpha$, then algebraically inverting the inequality to isolate θ gives an exact CI.

How to spot a pivot. If θ and the data always appear in the pdf in a specific combination—like $t - \theta$ or t/θ—that combination is the pivot.

3.2 Location Family Pivots

If $X_1, \ldots, X_n$ are iid from a location family $f(x - \mu)$, then $\bar{X} - \mu$ has a distribution free of μ. **Recipe: subtract the location parameter.**

Deriving the Z-interval. Let $X_i \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$ with σ^2 known. First, compute the variance of the sample mean. Since $\bar{X} = \frac{1}{n}\sum_{i=1}^n X_i$ and the X_i are independent:

$$\mathrm{Var}\Big(\bar{X}\Big) = \mathrm{Var}\left(\frac{1}{n}\sum_{i=1}^n X_i\right) = \frac{1}{n^2}\sum_{i=1}^n \mathrm{Var}(X_i) = \frac{1}{n^2} \cdot n\sigma^2 = \frac{\sigma^2}{n}.$$

The $1/n$ comes out as $1/n^2$ because variance scales as the square of constants; the sum of n independent terms contributes $n\sigma^2$. So $\bar{X} \sim \mathrm{n}(\mu,\, \sigma^2/n)$. Standardizing (subtract the mean, divide by the standard deviation):

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \sim \mathrm{n}(0, 1).$$

This Z is a pivot: it involves the unknown μ, but its distribution is completely known. Bounding: $\mathbb{P}\Big(-z_{\alpha/2} \leq Z \leq z_{\alpha/2}\Big) = 1 - \alpha$. Inverting:

$$\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \;\leq\; \mu \;\leq\; \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}.$$

Deriving the t-interval. When σ^2 is unknown, replace σ with S. The pivot becomes $T = (\bar{X} - \mu)/(S/\sqrt{n})$. To see that $T \sim t_{n-1}$, multiply and divide the numerator by σ:

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}} = \frac{(\bar{X} - \mu)/(\sigma/\sqrt{n})}{S/\sigma}.$$

Now rewrite the denominator: $S/\sigma = \sqrt{S^2/\sigma^2} = \sqrt{[(n-1)S^2/\sigma^2]/(n-1)}$. Defining $Z = (\bar{X} - \mu)/(\sigma/\sqrt{n})$ and $V = (n-1)S^2/\sigma^2$, this becomes

$$T = \frac{Z}{\sqrt{V/(n-1)}}.$$

We know $Z \sim \mathrm{n}(0,1)$ and $V \sim \chi^2_{n-1}$, and they are independent (a key normal-theory fact; see below). By definition, $Z/\sqrt{V/\nu}$ with $Z \perp V$ is t_ν, so $T \sim t_{n-1}$. The crucial point: the unknown σ appears in both Z and V, but it cancels in the ratio, leaving a parameter-free distribution. The CI is $\bar{X} \pm t_{n-1,\alpha/2} \cdot S/\sqrt{n}$.

Warning: z vs. t: When Does It Matter?

Use $z_{\alpha/2}$ only when σ is *known*. If σ is estimated by S, you must use $t_{n-1,\alpha/2}$. For $n < 30$, using z instead of t undercovers.

3.3 Scale Family Pivots

If $X_1, \ldots, X_n$ are iid from a scale family $\sigma^{-1} f(x/\sigma)$, then T/σ is a pivot. **Recipe: divide by the scale parameter.**

For a location-scale family $\sigma^{-1} f((x-\mu)/\sigma)$: use $(\bar{X} - \mu)/S$.

Exponential CI. Let $X_i \overset{\text{iid}}{\sim} \mathrm{Exp}(\lambda)$ with $\mathbb{E}[X_i] = \lambda$. The sufficient statistic $T = \sum X_i \sim \mathrm{Gamma}(n, \lambda)$. Since t and λ appear as t/λ in the pdf, define the pivot $Q = 2T/\lambda \sim \chi^2_{2n}$. Bounding: $\mathbb{P}\left(\chi^2_{2n,1-\alpha/2} \le Q \le \chi^2_{2n,\alpha/2}\right) = 1 - \alpha$. Since Q is decreasing in λ, inverting flips the direction:

$$\frac{2\sum X_i}{\chi^2_{2n,\alpha/2}} \le \lambda \le \frac{2\sum X_i}{\chi^2_{2n,1-\alpha/2}}.$$

Normal variance CI. $Q = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ is a scale-family pivot. Inverting gives $\frac{(n-1)S^2}{\chi^2_{n-1,\alpha/2}} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{n-1,1-\alpha/2}}$.

Warning: Unequal Tail Split for χ^2 Intervals

The χ^2 distribution is skewed, so equal tail splits do not give the shortest CI. The shortest satisfies $f(a) = f(b)$ (equal pdf heights). In practice, equal-tail is standard.

3.4 Location-Scale Pivot Summary

Family type	Form of pdf	Pivotal quantity
Location	$f(x-\mu)$	$\bar{X} - \mu$
Scale	$\sigma^{-1} f(x/\sigma)$	T/σ
Location-scale	$\sigma^{-1} f((x-\mu)/\sigma)$	$(\bar{X} - \mu)/S$

Quick rule: location $\Rightarrow$ subtract; scale $\Rightarrow$ divide; both $\Rightarrow$ subtract and divide.

3.5 Key Distribution Results for Pivots

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$. The following facts appear constantly:

1. $\bar{X} \sim \mathrm{n}(\mu, \sigma^2/n)$, so $(\bar{X} - \mu)/(\sigma/\sqrt{n}) \sim \mathrm{n}(0,1)$.
2. $(n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$. To see why: write $\sum(X_i - \bar{X})^2/\sigma^2 = \sum[(X_i - \mu)/\sigma - (\bar{X} - \mu)/\sigma]^2$. This is a sum of n standard normals ($Z_i = (X_i - \mu)/\sigma$) projected onto the $(n-1)$-dimensional subspace orthogonal to $\mathbf{1}$. The rank of the projection is $n-1$, so the result is χ^2_{n-1}.
3. $\bar{X}$ and S^2 are **independent**. Since S^2 is a function of the residuals $X_i - \bar{X}$, it suffices to show $\bar{X}$ and $X_i - \bar{X}$ are independent. For jointly normal random variables, zero covariance implies independence, so compute:

   $$\mathrm{Cov}\Big(\bar{X},\, X_i - \bar{X}\Big) = \mathrm{Cov}\Big(\bar{X},\, X_i\Big) - \mathrm{Var}\Big(\bar{X}\Big).$$

   For the first term, $\mathrm{Cov}\Big(\bar{X},\, X_i\Big) = \mathrm{Cov}\Big(\frac{1}{n}\sum_{j=1}^n X_j,\, X_i\Big) = \frac{1}{n}\,\mathrm{Var}(X_i) = \frac{\sigma^2}{n}$, since the X_j with $j \neq i$ are independent of X_i. The second term is $\mathrm{Var}\Big(\bar{X}\Big) = \sigma^2/n$. So

   $$\mathrm{Cov}\Big(\bar{X},\, X_i - \bar{X}\Big) = \frac{\sigma^2}{n} - \frac{\sigma^2}{n} = 0.$$

   Zero covariance plus joint normality gives independence. Since $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ is a function of the $(X_i - \bar{X})$, it is independent of $\bar{X}$.
4. $(\bar{X} - \mu)/(S/\sqrt{n}) \sim t_{n-1}$ (follows from 1–3 and the definition $t_\nu = Z/\sqrt{V/\nu}$).
5. For independent samples: $\frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{n-1,m-1}$.

Gamma–chi-squared connection. If $X_i \overset{\text{iid}}{\sim} \mathrm{Exp}(\lambda)$ then $2\sum X_i/\lambda \sim \chi^2_{2n}$. More generally, $T \sim \mathrm{Gamma}(\alpha, \beta)$ implies $2T/\beta \sim \chi^2_{2\alpha}$.

t-distribution. $T = Z/\sqrt{V/\nu}$ where $Z \sim \mathrm{n}(0,1)$, $V \sim \chi^2_\nu$, $Z \perp V$. Properties: $\mathbb{E}[T] = 0$ ($\nu > 1$), $\mathrm{Var}(T) = \nu/(\nu - 2)$ ($\nu > 2$), heavier tails than $\mathrm{n}(0,1)$, $t_\nu \to \mathrm{n}(0,1)$ as $\nu \to \infty$.

3.6 CI–Test Duality

For each θ_0, let $A(\theta_0)$ be the acceptance region of a level α test of $H_0\colon \theta = \theta_0$. Then $C(\mathbf{x}) = \{\theta_0 : \mathbf{x} \in A(\theta_0)\}$ is a $1-\alpha$ confidence set, and conversely.

In plain language: θ_0 is in the CI $\iff$ the test of $H_0\colon \theta = \theta_0$ fails to reject. A UMP test inverts to a uniformly most accurate CI; an unbiased test inverts to an unbiased CI.

3.7 The Pivot Recipe

1. Find a sufficient statistic T for θ.
2. Construct a pivot $Q(T, \theta)$ with a known, parameter-free distribution.
3. Choose quantiles a, b such that $\mathbb{P}(a \leq Q \leq b) = 1 - \alpha$.
4. Invert the inequality to isolate θ.

Optimizing interval length. Among all $[a, b]$ with $\int_a^b f(x)\,dx = 1-\alpha$, the shortest has $f(a) = f(b)$ (equal pdf heights at endpoints). For symmetric unimodal distributions this is the equal-tail split; for skewed distributions like χ^2 it is not.

4 Linear Regression — Tying It All Together

Regression is the application layer: estimation, testing, and intervals in one framework.

4.1 The Linear Model and OLS Derivation

The model is $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$, where $\mathbf{X}$ is $n \times p$ with rank p, $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$, and $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2 \mathbf{I}_n$.

Ordinary least squares (OLS) derivation. Minimize the residual sum of squares $\mathrm{RSS}(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$. Expanding:

$$\mathrm{RSS} = \mathbf{Y}^T\mathbf{Y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{Y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}.$$

Differentiating with respect to $\boldsymbol{\beta}$ and setting to zero:

$$\frac{\partial\,\mathrm{RSS}}{\partial\boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{Y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0}.$$

This gives the **normal equations** $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{Y}$. Since $\mathbf{X}$ has full column rank, $\mathbf{X}^T\mathbf{X}$ is invertible:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}.$$

Covariance form. The matrix formula has a clean interpretation in terms of covariances. Define the sample covariance matrix of the regressors and the sample covariance vector with Y:

$$S_{XX} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(\mathbf{x}_i - \bar{\mathbf{x}})^T, \qquad S_{XY} = \frac{1}{n}\sum_{i=1}^{n}(\mathbf{x}_i - \bar{\mathbf{x}})(Y_i - \bar{Y}).$$

When the model includes an intercept (as is typical), the slope coefficients satisfy

$$\hat{\boldsymbol{\beta}}_{\text{slopes}} = S_{XX}^{-1}\,S_{XY}.$$

This is the multivariate version of "covariance with Y divided by variance of X."

For **simple regression** ($Y_i = \beta_0 + \beta_1 x_i + \varepsilon_i$), S_{XX} and S_{XY} are scalars, and this reduces to

$$\hat{\beta}_1 = \frac{S_{XY}}{S_{XX}}, \qquad \hat{\beta}_0 = \bar{Y} - \hat{\beta}_1\bar{x}.$$

The intercept is determined by the constraint that the fitted line passes through $(\bar{x}, \bar{Y})$.

Geometric view. The fitted values are $\hat{\mathbf{Y}} = \mathbf{H}\mathbf{Y}$ where $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is the **hat matrix** (projection onto the column space of $\mathbf{X}$). The residuals are $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}$. Both $\mathbf{H}$ and $\mathbf{I}_n - \mathbf{H}$ are symmetric and idempotent, with $\mathrm{tr}(\mathbf{H}) = p$ and $\mathrm{tr}(\mathbf{I}_n - \mathbf{H}) = n - p$.

A key consequence of $\mathbf{H}$ being a projection: $\mathbf{X}^T\mathbf{e} = \mathbf{X}^T(\mathbf{I} - \mathbf{H})\mathbf{Y} = \mathbf{0}$, so residuals are orthogonal to all regressors.

Warning: The "Hat" Trick

h_{ii} measures **leverage** of observation i. $0 \le h_{ii} \le 1$, $\sum h_{ii} = p$, average $= p/n$. Flag points with $h_{ii} > 2p/n$.

4.2 Properties of OLS

Unbiasedness. $\mathbb{E}\left[\hat{\boldsymbol{\beta}}\right] = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}[\mathbf{Y}] = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$.

Variance. Since $\hat{\boldsymbol{\beta}} = \mathbf{AY}$ where $\mathbf{A} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ is a constant matrix, use the linear transformation rule $\mathrm{Var}(\mathbf{AY}) = \mathbf{A}\ \mathrm{Var}(\mathbf{Y})\ \mathbf{A}^T$:

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \mathbf{A} \cdot \sigma^2\mathbf{I}_n \cdot \mathbf{A}^T = \sigma^2\,(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

The middle step simplifies because $\mathbf{X}^T\mathbf{X}$ sandwiched between two $(\mathbf{X}^T\mathbf{X})^{-1}$ factors cancels one of them. Individual coefficient: $\mathrm{Var}\left(\hat{\beta}_j\right) = \sigma^2 d_{jj}$ where d_{jj} is the appropriate diagonal element of $(\mathbf{X}^T\mathbf{X})^{-1}$.

Covariance form. In the general case, $\mathrm{Var}\left(\hat{\boldsymbol{\beta}}_{\text{slopes}}\right) = \sigma^2\, S_{XX}^{-1}/n$: the covariance matrix of the slope estimates is proportional to the inverse of the regressor covariance matrix.

For **simple regression**, $\hat{\beta}_1 = \sum c_i Y_i$ where $c_i = (x_i - \bar{x})/(nS_{XX})$ are constants. Since the Y_i are independent with variance σ^2:

$$\mathrm{Var}\left(\hat{\beta}_1\right) = \sigma^2 \sum c_i^2 = \frac{\sigma^2\ \sum(x_i - \bar{x})^2}{(nS_{XX})^2} = \frac{\sigma^2\, nS_{XX}}{n^2 S_{XX}^2} = \frac{\sigma^2}{n\, S_{XX}}.$$

This makes the intuition transparent: the slope estimate is more precise when (1) noise σ^2 is small, (2) the sample size n is large, or (3) the regressor has high spread S_{XX} (the x-values are well spread out).

Gauss–Markov Theorem. Among all linear unbiased estimators of $\mathbf{a}^T\boldsymbol{\beta}$, the OLS estimator $\mathbf{a}^T\hat{\boldsymbol{\beta}}$ has the smallest variance (best linear unbiased estimator, or BLUE).

Proof sketch. Let $\tilde{\boldsymbol{\beta}} = \mathbf{CY}$ be any other linear unbiased estimator. Write $\mathbf{C} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \mathbf{D}$. Unbiasedness requires $\mathbf{CX} = \mathbf{I}$, which forces $\mathbf{DX} = \mathbf{0}$. Then:

$$\mathrm{Var}\left(\tilde{\boldsymbol{\beta}}\right) = \sigma^2\mathbf{CC}^T = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} + \sigma^2\mathbf{DD}^T.$$

Since $\mathbf{DD}^T$ is positive semidefinite, $\mathrm{Var}\left(\tilde{\boldsymbol{\beta}}\right) \geq \mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right)$ in the matrix ordering. □

Unbiased estimator of σ^2. $S^2 = \mathrm{RSS}/(n-p)$ where $\mathrm{RSS} = \mathbf{Y}^T(\mathbf{I}-\mathbf{H})\mathbf{Y}$. To verify $\mathbb{E}\left[S^2\right] = \sigma^2$, we need $\mathbb{E}[\mathrm{RSS}]$. Use the trace identity: for any symmetric matrix $\mathbf{M}$ and random vector $\mathbf{Z}$ with $\mathbb{E}[\mathbf{Z}] = \mathbf{0}$ and $\mathrm{Var}(\mathbf{Z}) = \boldsymbol{\Sigma}$,

$$\mathbb{E}\left[\mathbf{Z}^T\mathbf{MZ}\right] = \mathrm{tr}(\mathbf{M}\boldsymbol{\Sigma}).$$

(This follows from $\mathbb{E}\left[\mathbf{Z}^T\mathbf{MZ}\right] = \mathbb{E}\left[\mathrm{tr}(\mathbf{MZZ}^T)\right] = \mathrm{tr}(\mathbf{M}\,\mathbb{E}\left[\mathbf{ZZ}^T\right]) = \mathrm{tr}(\mathbf{M}\boldsymbol{\Sigma})$, using the fact that trace and expectation commute.) Applying this with $\mathbf{Z} = \mathbf{Y} - \mathbf{X}\boldsymbol{\beta} = \boldsymbol{\varepsilon}$, $\mathbf{M} = \mathbf{I} - \mathbf{H}$, and $\boldsymbol{\Sigma} = \sigma^2\mathbf{I}$:

$$\mathbb{E}[\mathrm{RSS}] = \mathrm{tr}[(\mathbf{I}-\mathbf{H})\,\sigma^2\mathbf{I}] = \sigma^2\,\mathrm{tr}(\mathbf{I}-\mathbf{H}) = \sigma^2(n-p).$$

Dividing by $n-p$ gives $\mathbb{E}\left[S^2\right] = \sigma^2$.

Warning: Dividing by $n-p$, Not n

We lose p degrees of freedom for estimating p parameters. For simple regression ($p=2$): divide by $n-2$, not $n-1$. The MLE $\hat{\sigma}^2_{\text{MLE}} = \text{RSS}/n$ is biased downward.

4.3 Distribution Theory Under Normality

Adding $\boldsymbol{\varepsilon} \sim N_n(\mathbf{0}, \sigma^2\mathbf{I}_n)$ gives exact distributional results:

(i) $\hat{\boldsymbol{\beta}} \sim N_p(\boldsymbol{\beta},\, \sigma^2(\mathbf{X}^T\mathbf{X})^{-1})$.

(ii) $\text{RSS}/\sigma^2 \sim \chi^2_{n-p}$.

(iii) $\hat{\boldsymbol{\beta}}$ and S^2 are **independent** (because $\hat{\boldsymbol{\beta}}$ depends on $\mathbf{Y}$ through $\mathbf{HY}$, RSS depends through $(\mathbf{I}-\mathbf{H})\mathbf{Y}$, and $\mathbf{H}(\mathbf{I}-\mathbf{H}) = \mathbf{0}$).

The independence is what lets us form t-statistics:

$$\frac{\hat{\beta}_j - \beta_j}{S\sqrt{d_{jj}}} = \frac{(\hat{\beta}_j - \beta_j)/(\sigma\sqrt{d_{jj}})}{\sqrt{S^2/\sigma^2}} = \frac{Z}{\sqrt{V/(n-p)}} \sim t_{n-p},$$

where $Z \sim \text{n}(0,1)$ and $V = \text{RSS}/\sigma^2 \sim \chi^2_{n-p}$ are independent.

4.4 Hypothesis Testing

t-test for individual coefficients. To test $H_0\colon \beta_j = \beta_{j,0}$: $T = (\hat{\beta}_j - \beta_{j,0})/(S\sqrt{d_{jj}}) \sim t_{n-p}$ under H_0. Reject if $|T| > t^{\alpha/2}_{n-p}$. Most common case: $\beta_{j,0} = 0$ (is regressor j significant?).

F-test for multiple constraints. To test $H_0\colon \mathbf{A}\boldsymbol{\beta} = \mathbf{c}$ with q constraints:

$$F = \frac{(\text{RSS}_H - \text{RSS})/q}{\text{RSS}/(n-p)} = \frac{(\text{RSS}_H - \text{RSS})/q}{S^2} \sim F_{q,\,n-p} \quad \text{under } H_0.$$

When $q=1$, $F = T^2$ and $F_{1,n-p} = t^2_{n-p}$ (the t-test is a special case).

Overall F-test and R^2. $R^2 = 1 - \text{RSS}/\text{TSS}$, where $\text{TSS} = \sum(Y_i - \bar{Y})^2 = nS_{YY}$ is the total sum of squares. In simple regression, $R^2 = r^2_{XY}$, the square of the sample correlation $r_{XY} = S_{XY}/\sqrt{S_{XX}\,S_{YY}}$. Testing H_0: all slopes zero:

$$F = \frac{R^2/(p-1)}{(1-R^2)/(n-p)} \sim F_{p-1,\,n-p}.$$

Warning: R^2 Always Increases When You Add Variables

Adding a regressor can never decrease R^2. Use adjusted $R^2 = 1 - \frac{n-1}{n-p}(1-R^2)$ or the F-test to decide if the improvement is significant. Under the null (all slopes zero), $\mathbb{E}[R^2] = (p-1)/(n-1)$, so even a useless model with 10 regressors and $n=100$ has $\mathbb{E}[R^2] \approx 0.09$.

4.5 Confidence and Prediction Intervals

Let $v_0 = \mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$ at a prediction point $\mathbf{x}_0$. In simple regression this reduces to

$$v_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{n\,S_{XX}}.$$

The first term is the baseline uncertainty from estimating the intercept; the second grows as x_0 moves away from the center of the data.

CI for β_j. $\hat{\beta}_j \pm t_{n-p}^{\alpha/2} \cdot S\sqrt{d_{jj}}$.

CI for mean response $\mathbb{E}[Y|\mathbf{x}_0]$. $\hat{Y}_0 \pm t_{n-p}^{\alpha/2} \cdot S\sqrt{v_0}$.

Prediction interval (PI) for a new Y_0. A new observation $Y_0 = \mathbf{x}_0^T\boldsymbol{\beta} + \varepsilon_0$ is independent of the training data, so

$$\mathrm{Var}\Big(\hat{Y}_0 - Y_0\Big) = \mathrm{Var}\Big(\hat{Y}_0\Big) + \mathrm{Var}(\varepsilon_0) = \sigma^2 v_0 + \sigma^2 = \sigma^2(v_0 + 1).$$

The PI is $\hat{Y}_0 \pm t_{n-p}^{\alpha/2} \cdot S\sqrt{v_0 + 1}$.

Warning: Confidence vs. Prediction Intervals

The CI uses $S\sqrt{v_0}$ (uncertainty in the mean); the PI uses $S\sqrt{v_0+1}$ (also captures irreducible noise). The PI is **always wider**. As $n \to \infty$, the CI shrinks to zero but the PI converges to $\pm z_{\alpha/2}\sigma$ (noise floor). In simple regression, both are narrowest at $x_0 = \bar{x}$ and widen as x_0 moves away—extrapolation is dangerous.

4.6 Key Interview Topics

Leverage and influence. $\hat{Y}_i = h_{ii}Y_i + \sum_{j\neq i} h_{ij}Y_j$, so $\mathrm{Var}(e_i) = \sigma^2(1 - h_{ii})$. High-leverage points have small residual variance, making outliers harder to detect from raw residuals. Use studentized residuals: $r_i = e_i/(S\sqrt{1 - h_{ii}})$.

Multicollinearity. When regressors are highly correlated, $\mathbf{X}^T\mathbf{X}$ is near-singular and $(\mathbf{X}^T\mathbf{X})^{-1}$ has large diagonal entries, inflating $\mathrm{Var}\Big(\hat{\beta}_j\Big)$. The variance inflation factor $\mathrm{VIF}_j = 1/(1 - R_j^2)$ measures this; values above 5–10 signal trouble. Symptoms: individual t-tests non-significant despite a significant overall F; unstable coefficient estimates.

Adding variables. $\mathrm{Var}\Big(\hat{Y}_{0,\text{enlarged}}\Big) \geq \mathrm{Var}\Big(\hat{Y}_{0,\text{original}}\Big)$. Adding irrelevant regressors always increases prediction variance even though it decreases training RSS. This is the bias–variance tradeoff.

Part 2: Worked Problems & Drills

Repeat these until they are automatic.

5 Point Estimation — Problems

5.1 Method of Moments

Example: MOM for Normal(μ, σ^2)

Let $X_1, \ldots, X_n \sim$ iid $N(\mu, \sigma^2)$, with both parameters unknown. We need two equations ($k = 2$).
Step 1: Compute population moments.

$$\mu_1' = \mathbb{E}[X] = \mu, \qquad \mu_2' = \mathbb{E}\left[X^2\right] = \mathrm{Var}(X) + (\mathbb{E}[X])^2 = \sigma^2 + \mu^2.$$

Step 2: Set sample moments equal to population moments.

$$m_1 = \bar{X} = \mu, \qquad m_2 = \frac{1}{n}\sum_{i=1}^{n} X_i^2 = \sigma^2 + \mu^2.$$

Step 3: Solve.
From the first equation:

$$\hat{\mu}_{\mathrm{MOM}} = \bar{X}.$$

Substituting into the second:

$$\hat{\sigma}^2_{\mathrm{MOM}} = \frac{1}{n}\sum_{i=1}^{n} X_i^2 - \bar{X}^2 = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^2.$$

Note: the MOM estimator of σ^2 divides by n, not $n-1$. It is biased: $\mathbb{E}[\hat{\sigma}^2_{\mathrm{MOM}}] = \frac{n-1}{n}\sigma^2$.

Example: MOM for Uniform$(0, \theta)$

Let $X_1, \ldots, X_n \sim$ iid Uniform$(0, \theta)$, with $\theta > 0$ unknown. Only one parameter, so we need one equation.
Step 1: Population moment.

$$\mu_1' = \mathbb{E}[X] = \frac{\theta}{2}.$$

Step 2: Set equal.

$$\bar{X} = \frac{\theta}{2}.$$

Step 3: Solve.

$$\hat{\theta}_{\mathrm{MOM}} = 2\bar{X}.$$

Quick check: Is it unbiased?

$$\mathbb{E}\left[2\bar{X}\right] = 2 \cdot \frac{\theta}{2} = \theta. \quad \checkmark$$

Example: MOM for Gamma(α, β)

Let $X_1, \ldots, X_n \sim$ iid Gamma(α, β) with pdf $f(x|\alpha, \beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$, $x > 0$. Two unknown parameters, so we need two equations.

Step 1: Population moments.

$$\mu_1' = \mathbb{E}[X] = \alpha\beta, \qquad \mu_2' = \mathbb{E}\left[X^2\right] = \mathrm{Var}(X) + (\mathbb{E}[X])^2 = \alpha\beta^2 + \alpha^2\beta^2.$$

Step 2: Set equal.

$$\bar{X} = \alpha\beta, \qquad \frac{1}{n}\sum X_i^2 = \alpha\beta^2 + \alpha^2\beta^2.$$

Step 3: Solve.
From the second equation minus the square of the first:

$$\frac{1}{n}\sum X_i^2 - \bar{X}^2 = \alpha\beta^2.$$

The left side is $\hat{\sigma}^2_{\mathrm{MOM}} = \frac{1}{n}\sum(X_i - \bar{X})^2$. So:

$$\hat{\beta}_{\mathrm{MOM}} = \frac{\frac{1}{n}\sum(X_i - \bar{X})^2}{\bar{X}}, \qquad \hat{\alpha}_{\mathrm{MOM}} = \frac{\bar{X}}{\hat{\beta}_{\mathrm{MOM}}} = \frac{\bar{X}^2}{\frac{1}{n}\sum(X_i - \bar{X})^2}.$$

Example: MOM for Beta(α, β)

Let $X_1, \ldots, X_n \sim$ iid Beta(α, β). Two unknowns, so we need two equations.
Step 1: Population moments.

$$\mu_1' = \mathbb{E}[X] = \frac{\alpha}{\alpha+\beta}, \qquad \mathrm{Var}(X) = \frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}.$$

It is easier to work with $\mathbb{E}[X]$ and $\mathrm{Var}(X)$ directly. Define $\mu = \mathbb{E}[X]$ and $v = \mathrm{Var}(X)$. Then:

$$\mu = \frac{\alpha}{\alpha+\beta}, \qquad v = \frac{\mu(1-\mu)}{\alpha+\beta+1}.$$

Step 2: Set equal.

$$\bar{X} = \mu, \qquad \frac{1}{n}\sum(X_i - \bar{X})^2 = v.$$

Step 3: Solve.
From the variance relation:

$$\alpha + \beta + 1 = \frac{\bar{X}(1-\bar{X})}{\hat{v}}, \qquad \text{so } \alpha + \beta = \frac{\bar{X}(1-\bar{X})}{\hat{v}} - 1.$$

Combined with $\alpha = \bar{X}(\alpha+\beta)$:

$$\hat{\alpha}_{\mathrm{MOM}} = \bar{X}\left(\frac{\bar{X}(1-\bar{X})}{\hat{v}} - 1\right), \qquad \hat{\beta}_{\mathrm{MOM}} = (1-\bar{X})\left(\frac{\bar{X}(1-\bar{X})}{\hat{v}} - 1\right),$$

where $\hat{v} = \frac{1}{n}\sum_{i=1}^n (X_i - \bar{X})^2$.

5.2 Maximum Likelihood Estimation

Example: MLE for Normal(μ, σ^2) – both unknown

Let $X_1, \ldots, X_n \sim$ iid $N(\mu, \sigma^2)$.
Step 1: Log-likelihood.

$$\ell(\mu, \sigma^2) = -\frac{n}{2}\log(2\pi) - \frac{n}{2}\log\sigma^2 - \frac{1}{2\sigma^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

Step 2: Differentiate with respect to μ.

$$\frac{\partial \ell}{\partial \mu} = \frac{1}{\sigma^2}\sum_{i=1}^{n}(x_i - \mu) = 0.$$

Solving:

$$\hat{\mu}_{\text{MLE}} = \bar{x}.$$

Step 3: Differentiate with respect to σ^2.

$$\frac{\partial \ell}{\partial \sigma^2} = -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4}\sum_{i=1}^{n}(x_i - \mu)^2 = 0.$$

Solving:

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}(x_i - \bar{x})^2.$$

Note: The MLE $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$ divides by n and is biased. The unbiased version $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ is *not* the MLE.

Example: MLE for Exponential(λ)

Let $X_1, \ldots, X_n \sim$ iid Exp(λ) with pdf $f(x|\lambda) = \lambda e^{-\lambda x}$, $x > 0$.
Log-likelihood:

$$\ell(\lambda) = n\log\lambda - \lambda\sum_{i=1}^{n} x_i.$$

Differentiate:

$$\frac{d\ell}{d\lambda} = \frac{n}{\lambda} - \sum_{i=1}^{n} x_i = 0.$$

Solve:

$$\hat{\lambda}_{\text{MLE}} = \frac{n}{\sum_{i=1}^{n} x_i} = \frac{1}{\bar{x}}.$$

Verify: $\frac{d^2\ell}{d\lambda^2} = -\frac{n}{\lambda^2} < 0$, confirming a maximum.
Alternative parametrization. If $f(x|\theta) = \frac{1}{\theta}e^{-x/\theta}$ (mean parametrization), then $\hat{\theta}_{\text{MLE}} = \bar{X}$.

Example: MLE for Poisson(λ)

Let $X_1, \ldots, X_n \sim$ iid Poisson(λ) with pmf $P(X = x|\lambda) = \frac{e^{-\lambda}\lambda^x}{x!}$, $x = 0, 1, 2, \ldots$
Log-likelihood:

$$\ell(\lambda) = -n\lambda + \left(\sum_{i=1}^{n} x_i\right) \log \lambda - \sum_{i=1}^{n} \log(x_i!).$$

Differentiate:

$$\frac{d\ell}{d\lambda} = -n + \frac{\sum x_i}{\lambda} = 0.$$

Solve:

$$\hat{\lambda}_{\text{MLE}} = \frac{\sum x_i}{n} = \bar{x}.$$

Verify: $\frac{d^2\ell}{d\lambda^2} = -\frac{\sum x_i}{\lambda^2} < 0$ (assuming at least one $x_i > 0$).

5.3 Bias, Variance, and MSE

Example: MSE of Sample Variance S^2 and MLE $\hat{\sigma}^2$

Let $X_1, \ldots, X_n \sim$ iid $N(\mu, \sigma^2)$. Two competing estimators of σ^2:
Estimator 1: $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ (unbiased).

$$\mathbb{E}\left[S^2\right] = \sigma^2, \qquad \text{Var}\left(S^2\right) = \frac{2\sigma^4}{n-1}.$$

$$\text{MSE}(S^2) = \text{Var}\left(S^2\right) = \frac{2\sigma^4}{n-1}.$$

Estimator 2: $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2 = \frac{n-1}{n}S^2$ (MLE, biased).

$$\mathbb{E}\left[\hat{\sigma}^2\right] = \frac{n-1}{n}\sigma^2, \qquad \text{Bias} = -\frac{\sigma^2}{n}.$$

$$\text{Var}\left(\hat{\sigma}^2\right) = \left(\frac{n-1}{n}\right)^2 \text{Var}\left(S^2\right) = \frac{(n-1)^2}{n^2} \cdot \frac{2\sigma^4}{n-1} = \frac{2(n-1)\sigma^4}{n^2}.$$

$$\text{MSE}(\hat{\sigma}^2) = \frac{2(n-1)\sigma^4}{n^2} + \frac{\sigma^4}{n^2} = \frac{2n-1}{n^2}\sigma^4.$$

Comparison:

$$\text{MSE}(\hat{\sigma}^2) = \frac{2n-1}{n^2}\sigma^4 < \frac{2}{n-1}\sigma^4 = \text{MSE}(S^2).$$

The biased MLE has **smaller MSE** than the unbiased S^2. By trading a small amount of bias for a larger reduction in variance, the MLE wins.
Even better: The estimator $\tilde{S}^2 = \frac{1}{n+1}\sum(X_i - \bar{X})^2$ minimizes MSE among all estimators of the form $b \cdot S^2$ under normality.

Example: MSE Comparison: MLE vs. MOM for Uniform$(0, \theta)$

We derived $\hat{\theta}_{\text{MLE}} = X_{(n)}$ and $\hat{\theta}_{\text{MOM}} = 2\bar{X}$.
Analysis of the MLE, $X_{(n)}$:
The pdf of $Y = X_{(n)} = \max(X_1, \ldots, X_n)$ for Uniform$(0, \theta)$ is:

$$f_Y(y) = \frac{ny^{n-1}}{\theta^n}, \quad 0 < y < \theta.$$

Computing moments:

$$\mathbb{E}\Big[X_{(n)}\Big] = \int_0^\theta y \cdot \frac{ny^{n-1}}{\theta^n}\, dy = \frac{n}{n+1}\theta.$$

$$\text{Bias}(X_{(n)}) = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}.$$

$$\mathbb{E}\Big[X_{(n)}^2\Big] = \int_0^\theta y^2 \cdot \frac{ny^{n-1}}{\theta^n}\, dy = \frac{n}{n+2}\theta^2.$$

$$\text{Var}\Big(X_{(n)}\Big) = \frac{n}{n+2}\theta^2 - \left(\frac{n}{n+1}\right)^2 \theta^2 = \frac{n\theta^2}{(n+1)^2(n+2)}.$$

$$\text{MSE}(X_{(n)}) = \text{Var}\Big(X_{(n)}\Big) + \text{Bias}^2 = \frac{n\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2} = \frac{2\theta^2}{(n+1)(n+2)}.$$

Note: The unbiased version is $\frac{n+1}{n}X_{(n)}$, with variance $\frac{\theta^2}{n(n+2)}$.
Analysis of the MOM, $2\bar{X}$:

$$\mathbb{E}\Big[2\bar{X}\Big] = 2 \cdot \frac{\theta}{2} = \theta \quad \text{(unbiased)}.$$

$$\text{Var}\Big(2\bar{X}\Big) = 4\,\text{Var}\Big(\bar{X}\Big) = \frac{4}{n}\,\text{Var}(X) = \frac{4}{n} \cdot \frac{\theta^2}{12} = \frac{\theta^2}{3n}.$$

$$\text{MSE}(2\bar{X}) = \frac{\theta^2}{3n}.$$

Comparison:

$$\frac{\text{MSE}(X_{(n)})}{\text{MSE}(2\bar{X})} = \frac{2\theta^2}{(n+1)(n+2)} \cdot \frac{3n}{\theta^2} = \frac{6n}{(n+1)(n+2)}.$$

For large n this ratio approaches $\frac{6}{n} \to 0$, so the MLE is **vastly superior** to the MOM in terms of MSE. For example, with $n = 10$: $\text{MSE}(X_{(n)})/\text{MSE}(2\bar{X}) \approx 0.45$. With $n = 100$: the ratio is approximately 0.058.

5.4 Sufficiency and the Factorization Theorem

Example: Sufficiency for Exponential(λ)

Let $X_1, \ldots, X_n \sim$ iid Exp(λ), $f(x|\lambda) = \lambda e^{-\lambda x}$ for $x > 0$.
Joint density:

$$f(\mathbf{x}|\lambda) = \prod_{i=1}^n \lambda e^{-\lambda x_i} = \lambda^n \exp\left(-\lambda \sum_{i=1}^n x_i\right).$$

Factorization:

$$\underbrace{\lambda^n \exp\Big(-\lambda \sum x_i\Big)}_{g(T(\mathbf{x}),\,\lambda)} \cdot \underbrace{1}_{h(\mathbf{x})}, \qquad T(\mathbf{x}) = \sum_{i=1}^{n} x_i.$$

So $T = \sum X_i$ (equivalently $\bar{X}$) is sufficient for λ.

Example: Sufficiency for Normal(μ, σ^2)

Let $X_1, \ldots, X_n \sim$ iid $N(\mu, \sigma^2)$, both unknown.
Joint density:

$$f(\mathbf{x}|\mu, \sigma^2) = \frac{1}{(2\pi\sigma^2)^{n/2}} \exp\left(-\frac{1}{2\sigma^2} \sum_{i=1}^{n} (x_i - \mu)^2\right).$$

Expand the exponent:

$$\sum (x_i - \mu)^2 = \sum x_i^2 - 2\mu \sum x_i + n\mu^2.$$

So the joint density depends on the data only through $\sum x_i$ and $\sum x_i^2$.
Sufficient statistic: $T(\mathbf{X}) = (\sum X_i, \sum X_i^2)$, or equivalently $(\bar{X}, S^2)$.
When σ^2 is known and only μ is unknown, $T = \sum X_i$ alone suffices. When μ is known and only σ^2 is unknown, $T = \sum (X_i - \mu)^2$ suffices.

Example: Sufficiency for Uniform$(0, \theta)$

Let $X_1, \ldots, X_n \sim$ iid Uniform$(0, \theta)$.
Joint density:

$$f(\mathbf{x}|\theta) = \prod_{i=1}^{n} \frac{1}{\theta} \mathbf{1}_{(0,\theta)}(x_i) = \frac{1}{\theta^n} \prod_{i=1}^{n} \mathbf{1}_{(0,\theta)}(x_i).$$

The product of indicators equals 1 if and only if every $x_i \in (0, \theta)$. Since all observations are positive, this reduces to $x_{(n)} < \theta$:

$$f(\mathbf{x}|\theta) = \underbrace{\frac{1}{\theta^n} \mathbf{1}_{(0,\theta)}(x_{(n)})}_{g(x_{(n)},\,\theta)} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

Sufficient statistic: $T = X_{(n)} = \max(X_1, \ldots, X_n)$.
This is also a **complete** sufficient statistic.

5.5 Rao–Blackwell Theorem

Example: Rao–Blackwell for Uniform$(0, \theta)$

Goal: Find the best unbiased estimator of θ.
Step 1: Start with a crude unbiased estimator.
Take $W = 2X_1$ (just use the first observation).

$$\mathbb{E}[2X_1] = 2 \cdot \frac{\theta}{2} = \theta \quad \checkmark.$$

$$\mathrm{Var}(2X_1) = 4 \cdot \frac{\theta^2}{12} = \frac{\theta^2}{3}.$$

Step 2: Identify the sufficient statistic.
$T = X_{(n)} = \max(X_1, \ldots, X_n)$ is a complete sufficient statistic.
Step 3: Compute $\phi(T) = \mathbb{E}\Big[2X_1 \mid X_{(n)}\Big]$.
By symmetry, conditional on $X_{(n)} = t$, the observation X_1 either equals t (if X_1 is the maximum) or is uniform on $(0, t)$ (if X_1 is not the maximum).

$$\begin{aligned}\mathbb{E}\Big[X_1 \mid X_{(n)} = t\Big] &= \mathbb{P}\Big(X_1 = X_{(n)}\Big) \cdot \mathbb{E}\Big[X_1 \mid X_1 = X_{(n)},\, X_{(n)} = t\Big] \\ &\quad + \mathbb{P}\Big(X_1 \neq X_{(n)}\Big) \cdot \mathbb{E}\Big[X_1 \mid X_1 \neq X_{(n)},\, X_{(n)} = t\Big] \\ &= \frac{1}{n} \cdot t + \frac{n-1}{n} \cdot \frac{t}{2} = \frac{t}{n} + \frac{(n-1)t}{2n} = \frac{(n+1)t}{2n}.\end{aligned}$$

So:

$$\phi(T) = 2 \cdot \frac{(n+1)}{2n} X_{(n)} = \frac{n+1}{n} X_{(n)}.$$

Step 4: Verify.

$$\mathbb{E}\left[\frac{n+1}{n} X_{(n)}\right] = \frac{n+1}{n} \cdot \frac{n}{n+1}\theta = \theta. \quad \checkmark$$

$$\mathrm{Var}\left(\frac{n+1}{n} X_{(n)}\right) = \left(\frac{n+1}{n}\right)^2 \cdot \frac{n\theta^2}{(n+1)^2(n+2)} = \frac{\theta^2}{n(n+2)}.$$

Variance comparison:

$$\mathrm{Var}(2X_1) = \frac{\theta^2}{3}, \qquad \mathrm{Var}\left(\frac{n+1}{n} X_{(n)}\right) = \frac{\theta^2}{n(n+2)}.$$

For $n = 10$: variance reduced from $\theta^2/3 \approx 0.333\theta^2$ to $\theta^2/120 \approx 0.0083\theta^2$ – a factor of 40 improvement.
Since $X_{(n)}$ is *complete* sufficient, $\frac{n+1}{n} X_{(n)}$ is the **UMVUE** of θ.

5.6 Cramér–Rao Lower Bound

Example: Cramér–Rao for Poisson(λ)

$f(x|\lambda) = e^{-\lambda}\lambda^x / x!$.
Score function:

$$\frac{\partial}{\partial\lambda} \log f(X|\lambda) = \frac{\partial}{\partial\lambda}\left(-\lambda + X \log\lambda - \log X!\right) = -1 + \frac{X}{\lambda}.$$

Fisher information:

$$I(\lambda) = \mathbb{E}\left[\left(-1 + \frac{X}{\lambda}\right)^2\right] = \mathrm{Var}\left(\frac{X}{\lambda}\right) = \frac{1}{\lambda^2}\,\mathrm{Var}(X) = \frac{1}{\lambda^2} \cdot \lambda = \frac{1}{\lambda}.$$

Cramér–Rao bound: For any unbiased estimator W of λ (so $\tau(\lambda) = \lambda$, $\tau' = 1$):

$$\mathrm{Var}(W) \geq \frac{1}{n \cdot (1/\lambda)} = \frac{\lambda}{n}.$$

Since $\mathrm{Var}\left(\bar{X}\right) = \lambda/n$, the sample mean $\bar{X}$ **attains the bound** and is the best unbiased estimator of λ.

Example: Cramér–Rao for Normal Variance

Let $X_1, \ldots, X_n \sim$ iid $N(\mu, \sigma^2)$ with μ known. We want to estimate σ^2.
Fisher information for σ^2:

$$\frac{\partial^2}{\partial(\sigma^2)^2} \log f(X|\mu, \sigma^2) = \frac{\partial^2}{\partial(\sigma^2)^2}\left(-\frac{1}{2}\log\sigma^2 - \frac{(X-\mu)^2}{2\sigma^2}\right) = \frac{1}{2\sigma^4} - \frac{(X-\mu)^2}{\sigma^6}.$$

$$I(\sigma^2) = -\mathbb{E}\left[\frac{1}{2\sigma^4} - \frac{(X-\mu)^2}{\sigma^6}\right] = -\frac{1}{2\sigma^4} + \frac{1}{\sigma^4} = \frac{1}{2\sigma^4}.$$

Cramér–Rao bound:

$$\mathrm{Var}(W) \geq \frac{1}{n/(2\sigma^4)} = \frac{2\sigma^4}{n}.$$

The estimator $\frac{1}{n}\sum(X_i - \mu)^2$ (with μ *known*) has variance $2\sigma^4/n$ and attains the bound. When μ is *unknown*, S^2 has $\mathrm{Var}(S^2) = 2\sigma^4/(n-1) > 2\sigma^4/n$, so S^2 does **not** attain the Cramér–Rao bound.

5.7 Drill Problems

Drill: MOM – Crank Through These

1. Let $X_1, \ldots, X_n \sim$ iid Uniform$(\theta, 2\theta)$, $\theta > 0$. Find the MOM estimator of θ.

   Solution. The first population moment is:

   $$\mathbb{E}[X] = \frac{\theta + 2\theta}{2} = \frac{3\theta}{2}.$$

   Setting the sample moment equal to the population moment:

   $$\bar{X} = \frac{3\theta}{2}.$$

   Solving for θ:

   $$\hat{\theta}_{\mathrm{MOM}} = \frac{2}{3}\bar{X}.$$

   Quick check: $\mathbb{E}\left[\frac{2}{3}\bar{X}\right] = \frac{2}{3} \cdot \frac{3\theta}{2} = \theta$, so the estimator is unbiased.

2. Let $X_1, \ldots, X_n \sim$ iid Geometric(p), $P(X = k) = (1-p)^{k-1}p$, $k = 1, 2, \ldots$ Find the MOM estimator of p.

   Solution. The first population moment is:

   $$\mathbb{E}[X] = \frac{1}{p}.$$

Setting the sample moment equal:

$$\bar{X} = \frac{1}{p}.$$

Solving for p:

$$\hat{p}_{\text{MOM}} = \frac{1}{\bar{X}}.$$

3. Let $X_1, \ldots, X_n \sim$ iid Pareto(α, x_m) with known x_m and pdf $f(x) = \alpha x_m^\alpha / x^{\alpha+1}$, $x > x_m$. Find the MOM estimator of α.

   Solution. The first population moment is:

$$\mathbb{E}[X] = \int_{x_m}^{\infty} x \cdot \frac{\alpha x_m^\alpha}{x^{\alpha+1}}\, dx = \alpha x_m^\alpha \int_{x_m}^{\infty} x^{-\alpha}\, dx = \alpha x_m^\alpha \cdot \frac{x_m^{1-\alpha}}{\alpha - 1} = \frac{\alpha x_m}{\alpha - 1},$$

   provided $\alpha > 1$.

   Setting the sample moment equal:

$$\bar{X} = \frac{\alpha x_m}{\alpha - 1}.$$

   Cross-multiplying: $\bar{X}(\alpha - 1) = \alpha x_m$, so $\alpha \bar{X} - \bar{X} = \alpha x_m$, giving $\alpha(\bar{X} - x_m) = \bar{X}$.

   Solving for α:

$$\hat{\alpha}_{\text{MOM}} = \frac{\bar{X}}{\bar{X} - x_m}.$$

Drill: MLE – Standard Calculus Approach

1. $X_1, \ldots, X_n \sim$ iid $N(\mu, \sigma_0^2)$ with σ_0^2 known. Find the MLE of μ. Verify the second derivative condition.

   Solution. The log-likelihood is:

$$\ell(\mu) = -\frac{n}{2}\log(2\pi\sigma_0^2) - \frac{1}{2\sigma_0^2}\sum_{i=1}^{n}(x_i - \mu)^2.$$

   Differentiating with respect to μ:

$$\frac{d\ell}{d\mu} = \frac{1}{\sigma_0^2}\sum_{i=1}^{n}(x_i - \mu) = \frac{1}{\sigma_0^2}(n\bar{x} - n\mu) = 0.$$

   Solving:

$$\hat{\mu}_{\text{MLE}} = \bar{X}.$$

   The second derivative is:

$$\frac{d^2\ell}{d\mu^2} = -\frac{n}{\sigma_0^2} < 0,$$

   confirming a maximum.

2. $X_1, \ldots, X_n \sim$ iid Bernoulli(p). Find the MLE of p.

   Solution. The log-likelihood is:

$$\ell(p) = \left(\sum_{i=1}^{n} x_i\right) \log p + \left(n - \sum_{i=1}^{n} x_i\right) \log(1-p).$$

   Differentiating:

$$\frac{d\ell}{dp} = \frac{\sum x_i}{p} - \frac{n - \sum x_i}{1-p} = 0.$$

   Cross-multiplying: $(1-p)\sum x_i = p(n - \sum x_i)$, which gives $\sum x_i = np$.

   Solving:

$$\hat{p}_{\text{MLE}} = \frac{\sum x_i}{n} = \bar{X}.$$

   The second derivative is $\frac{d^2\ell}{dp^2} = -\frac{\sum x_i}{p^2} - \frac{n - \sum x_i}{(1-p)^2} < 0$, confirming a maximum.

3. $X_1, \ldots, X_n \sim$ iid Gamma(α_0, β) with α_0 known. Find the MLE of β.

   Solution. The log-likelihood is:

$$\ell(\beta) = -n\alpha_0 \log \beta - \frac{\sum x_i}{\beta} + (\alpha_0 - 1)\sum \log x_i - n \log \Gamma(\alpha_0).$$

   Only the first two terms involve β. Differentiating:

$$\frac{d\ell}{d\beta} = -\frac{n\alpha_0}{\beta} + \frac{\sum x_i}{\beta^2} = 0.$$

   Multiplying through by β^2: $\sum x_i = n\alpha_0\beta$.

   Solving:

$$\hat{\beta}_{\text{MLE}} = \frac{\sum x_i}{n\alpha_0} = \frac{\bar{X}}{\alpha_0}.$$

   The second derivative is $\frac{d^2\ell}{d\beta^2} = \frac{n\alpha_0}{\beta^2} - \frac{2\sum x_i}{\beta^3}$. At $\hat{\beta}$ this equals $-\frac{n\alpha_0}{\hat{\beta}^2} < 0$, confirming a maximum.

4. Use the invariance property: if $X_1, \ldots, X_n \sim N(\mu, \sigma^2)$, what is the MLE of $\mathbb{P}(X > 0) = \Phi(\mu/\sigma)$?

   Solution. The MLEs of the individual parameters are $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.

   The invariance property of MLEs states: if $\hat{\theta}$ is the MLE of θ, then $g(\hat{\theta})$ is the MLE of $g(\theta)$ for any function g.

   The quantity $\mathbb{P}(X > 0)$ is a function of (μ, σ):

$$\mathbb{P}(X > 0) = \mathbb{P}\left(\frac{X - \mu}{\sigma} > \frac{-\mu}{\sigma}\right) = \Phi\left(\frac{\mu}{\sigma}\right).$$

   Applying invariance:

$$\widehat{\mathbb{P}(X > 0)} = \Phi\left(\frac{\hat{\mu}}{\hat{\sigma}}\right) = \Phi\left(\frac{\bar{X}}{\sqrt{\frac{1}{n}\sum(X_i - \bar{X})^2}}\right).$$

Drill: MLE – The Indicator Function Trap

1. $X_1, \ldots, X_n \sim$ iid Uniform$(\theta, \theta+1)$, $\theta \in \mathbb{R}$. Write the likelihood. Show that the MLE is any value in $[X_{(n)} - 1,\ X_{(1)}]$.

   Solution. Each observation has pdf $f(x_i|\theta) = \mathbf{1}_{(\theta,\,\theta+1)}(x_i)$. The joint likelihood is:

   $$L(\theta) = \prod_{i=1}^{n} \mathbf{1}_{(\theta,\,\theta+1)}(x_i).$$

   The product of indicators equals 1 if and only if every observation satisfies $\theta < x_i < \theta + 1$, which requires:

   $$\theta \le x_{(1)} \quad \text{and} \quad x_{(n)} \le \theta + 1.$$

   Rewriting: the likelihood equals 1 when $x_{(n)} - 1 \le \theta \le x_{(1)}$, and equals 0 otherwise.

   Since $L(\theta) = 1$ is constant (and maximal) on the entire interval $[x_{(n)} - 1,\ x_{(1)}]$, any value in this interval is an MLE. In particular, the MLE is not unique.

   Note that the interval is nonempty for valid data: since all observations lie in $(\theta, \theta+1)$, we have $x_{(n)} - x_{(1)} < 1$, i.e. $x_{(n)} - 1 < x_{(1)}$.

2. $X_1, \ldots, X_n \sim$ iid Uniform$(\theta, 2\theta)$, $\theta > 0$. Find the MLE of θ.

   Solution. Each observation has pdf $f(x_i|\theta) = \frac{1}{\theta}\mathbf{1}_{(\theta,\,2\theta)}(x_i)$. The joint likelihood is:

   $$L(\theta) = \prod_{i=1}^{n} \frac{1}{\theta}\mathbf{1}_{(\theta,\,2\theta)}(x_i) = \frac{1}{\theta^n}\prod_{i=1}^{n} \mathbf{1}_{(\theta,\,2\theta)}(x_i).$$

   The product of indicators equals 1 if and only if every observation satisfies $\theta < x_i < 2\theta$, which is equivalent to requiring:

   $$\theta \le x_{(1)} \quad \text{and} \quad x_{(n)} \le 2\theta.$$

   Rewriting, the likelihood is nonzero precisely when:

   $$\frac{x_{(n)}}{2} \le \theta \le x_{(1)}.$$

   On this interval the likelihood is $L(\theta) = 1/\theta^n$, which is strictly decreasing in θ (since $\theta > 0$ and $n \ge 1$).

   To maximize $L(\theta)$ we want θ as small as possible within the feasible set. The smallest feasible value is $\theta = x_{(n)}/2$, so:

   $$\hat{\theta}_{\text{MLE}} = \frac{X_{(n)}}{2}.$$

   However, we must also check that the feasible set is nonempty, i.e., that $x_{(n)}/2 \le x_{(1)}$. If $x_{(n)} > 2x_{(1)}$ the data are inconsistent with the model and no MLE exists. For valid data, $x_{(n)}/2 \le x_{(1)}$ always holds, and the MLE is $X_{(n)}/2$.

   More precisely, combining both boundary constraints: $\hat{\theta} = \max(X_{(n)}/2,\ X_{(1)})$. But since valid data from this model always satisfy $X_{(1)} \ge \theta$ and $X_{(n)} \le 2\theta$, we have

$X_{(1)} \geq \theta \geq X_{(n)}/2$, so the binding constraint is $\hat{\theta} = X_{(1)}$ when $X_{(1)} > X_{(n)}/2$, and $\hat{\theta} = X_{(n)}/2$ otherwise. In general:

$$\hat{\theta}_{\text{MLE}} = \max\left(\frac{X_{(n)}}{2},\ X_{(1)}\right).$$

3. $X_1, \ldots, X_n \sim$ iid with pdf $f(x|\theta) = \frac{2x}{\theta^2}$, $0 < x < \theta$. Find the MLE of θ.

   Solution. Each observation has pdf $f(x_i|\theta) = \frac{2x_i}{\theta^2}\mathbf{1}_{(0,\theta)}(x_i)$. The joint likelihood is:

$$L(\theta) = \prod_{i=1}^{n} \frac{2x_i}{\theta^2}\mathbf{1}_{(0,\theta)}(x_i) = \frac{2^n \prod_{i=1}^{n} x_i}{\theta^{2n}} \prod_{i=1}^{n} \mathbf{1}_{(0,\theta)}(x_i).$$

   The product of indicators equals 1 if and only if all observations satisfy $0 < x_i < \theta$. Since the data are positive, this reduces to $\theta \geq x_{(n)}$.

   On the feasible region $\theta \geq x_{(n)}$, the likelihood is:

$$L(\theta) = \frac{2^n \prod x_i}{\theta^{2n}},$$

   which is strictly decreasing in θ (since $2n \geq 2$). To maximize, we take θ as small as possible:

$$\hat{\theta}_{\text{MLE}} = X_{(n)}.$$

   Note that calculus does not apply here: differentiating the likelihood ignoring the indicator would give $\hat{\theta} = 0$, which is outside the parameter space. The support constraint is what drives the MLE to the boundary.

Drill: Bias and MSE Calculations

1. Let $X_1, \ldots, X_n \sim$ iid $\text{Exp}(1/\theta)$ (mean θ). The MLE is $\hat{\theta} = \bar{X}$. Compute the bias, variance, and MSE.

   Solution. Since each X_i has mean θ and variance θ^2 (the variance of an exponential with mean θ is θ^2), we compute:

$$\mathbb{E}\left[\bar{X}\right] = \mathbb{E}[X_1] = \theta.$$

   So the bias is:

$$\text{Bias}(\bar{X}) = \mathbb{E}\left[\bar{X}\right] - \theta = 0.$$

   For the variance:

$$\text{Var}\left(\bar{X}\right) = \frac{\text{Var}(X_1)}{n} = \frac{\theta^2}{n}.$$

   Since the estimator is unbiased, the MSE equals the variance:

$$\text{MSE}(\bar{X}) = \text{Var}\left(\bar{X}\right) + \text{Bias}^2 = \frac{\theta^2}{n} + 0 = \frac{\theta^2}{n}.$$

2. For the MLE of $\text{Uniform}(0, \theta)$, compute $\mathbb{E}\left[X_{(n)}\right]$, $\text{Var}\left(X_{(n)}\right)$, and $\text{MSE}(X_{(n)})$ from scratch.

Solution. The CDF of a single Uniform$(0,\theta)$ observation is $F(x) = x/\theta$ for $0 < x < \theta$. The CDF of the maximum $Y = X_{(n)}$ is:

$$F_Y(y) = \mathbb{P}\Big(X_{(n)} \le y\Big) = \mathbb{P}(X_1 \le y, \dots, X_n \le y) = \left(\frac{y}{\theta}\right)^n, \quad 0 < y < \theta.$$

Differentiating gives the pdf:

$$f_Y(y) = \frac{ny^{n-1}}{\theta^n}, \quad 0 < y < \theta.$$

Now compute the first moment:

$$\mathbb{E}\Big[X_{(n)}\Big] = \int_0^\theta y \cdot \frac{ny^{n-1}}{\theta^n}\, dy = \frac{n}{\theta^n}\int_0^\theta y^n\, dy = \frac{n}{\theta^n}\cdot\frac{\theta^{n+1}}{n+1} = \frac{n}{n+1}\theta.$$

The bias is:

$$\text{Bias}(X_{(n)}) = \frac{n}{n+1}\theta - \theta = -\frac{\theta}{n+1}.$$

For the variance, first compute the second moment:

$$\mathbb{E}\Big[X_{(n)}^2\Big] = \int_0^\theta y^2 \cdot \frac{ny^{n-1}}{\theta^n}\, dy = \frac{n}{\theta^n}\cdot\frac{\theta^{n+2}}{n+2} = \frac{n}{n+2}\theta^2.$$

Then:

$$\text{Var}\Big(X_{(n)}\Big) = \mathbb{E}\Big[X_{(n)}^2\Big] - \Big(\mathbb{E}\Big[X_{(n)}\Big]\Big)^2 = \frac{n}{n+2}\theta^2 - \frac{n^2}{(n+1)^2}\theta^2.$$

Combining over a common denominator:

$$\text{Var}\Big(X_{(n)}\Big) = \theta^2\left(\frac{n(n+1)^2 - n^2(n+2)}{(n+1)^2(n+2)}\right) = \theta^2 \cdot \frac{n}{(n+1)^2(n+2)}.$$

To verify the numerator simplification: $n(n+1)^2 - n^2(n+2) = n(n^2+2n+1) - n^2(n+2) = n^3 + 2n^2 + n - n^3 - 2n^2 = n$.

Finally, the MSE:

$$\text{MSE}(X_{(n)}) = \text{Var}\Big(X_{(n)}\Big) + \text{Bias}^2 = \frac{n\theta^2}{(n+1)^2(n+2)} + \frac{\theta^2}{(n+1)^2}.$$

$$= \frac{\theta^2}{(n+1)^2}\left(\frac{n}{n+2} + 1\right) = \frac{\theta^2}{(n+1)^2}\cdot\frac{2n+2}{n+2} = \frac{2\theta^2}{(n+1)(n+2)}.$$

3. Show that $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ is unbiased for σ^2 for *any* distribution with finite variance (not just normal).

   Solution. Let $\mu = \mathbb{E}[X_i]$ and $\sigma^2 = \text{Var}(X_i)$. Start by expanding the sum of squares:

$$\sum_{i=1}^n (X_i - \bar{X})^2 = \sum_{i=1}^n X_i^2 - n\bar{X}^2.$$

Now take expectations of each term. For the first:

$$\mathbb{E}\left[\sum_{i=1}^{n} X_i^2\right] = n\mathbb{E}\left[X_1^2\right] = n(\sigma^2 + \mu^2).$$

For the second, we use $\mathrm{Var}\left(\bar{X}\right) = \sigma^2/n$ and $\mathbb{E}\left[\bar{X}\right] = \mu$:

$$\mathbb{E}\left[n\bar{X}^2\right] = n\left(\mathrm{Var}\left(\bar{X}\right) + (\mathbb{E}\left[\bar{X}\right])^2\right) = n\left(\frac{\sigma^2}{n} + \mu^2\right) = \sigma^2 + n\mu^2.$$

Subtracting:

$$\mathbb{E}\left[\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = n(\sigma^2 + \mu^2) - (\sigma^2 + n\mu^2) = (n-1)\sigma^2.$$

Dividing both sides by $n-1$:

$$\mathbb{E}\left[S^2\right] = \mathbb{E}\left[\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2\right] = \sigma^2.$$

This derivation used only $\mathbb{E}[X_i] = \mu$, $\mathrm{Var}(X_i) = \sigma^2$, and independence (for the variance of $\bar{X}$). No normality assumption was needed.

Drill: Sufficiency – Apply the Factorization Theorem

For each distribution below, write the joint density, factor it, and identify the sufficient statistic.

1. $X_1, \ldots, X_n \sim$ iid Bernoulli(p).

   Solution. Each X_i takes values in $\{0, 1\}$ with pmf $f(x_i|p) = p^{x_i}(1-p)^{1-x_i}$. The joint pmf is:

   $$f(\mathbf{x}|p) = \prod_{i=1}^{n} p^{x_i}(1-p)^{1-x_i} = p^{\sum x_i}(1-p)^{n-\sum x_i}.$$

   Apply the factorization theorem:

   $$f(\mathbf{x}|p) = \underbrace{p^{\sum x_i}(1-p)^{n-\sum x_i}}_{g(T(\mathbf{x}),p)} \cdot \underbrace{1}_{h(\mathbf{x})}, \qquad T(\mathbf{x}) = \sum_{i=1}^{n} x_i.$$

   The joint pmf depends on the data only through $\sum x_i$, so $T = \sum X_i$ is sufficient for p.

2. $X_1, \ldots, X_n \sim$ iid $N(\mu, \sigma_0^2)$, σ_0^2 known.

   Solution. The joint density is:

   $$f(\mathbf{x}|\mu) = \prod_{i=1}^{n} \frac{1}{\sqrt{2\pi\sigma_0^2}} \exp\left(-\frac{(x_i-\mu)^2}{2\sigma_0^2}\right) = \frac{1}{(2\pi\sigma_0^2)^{n/2}} \exp\left(-\frac{1}{2\sigma_0^2}\sum_{i=1}^{n}(x_i-\mu)^2\right).$$

Expand the exponent:

$$\sum_{i=1}^{n}(x_i-\mu)^2 = \sum_{i=1}^{n} x_i^2 - 2\mu\sum_{i=1}^{n} x_i + n\mu^2.$$

Now factor:

$$f(\mathbf{x}|\mu) = \underbrace{\frac{1}{(2\pi\sigma_0^2)^{n/2}}\exp\left(-\frac{\sum x_i^2 - 2\mu\sum x_i + n\mu^2}{2\sigma_0^2}\right)}_{g(\sum x_i,\, \sum x_i^2,\, \mu)} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

Since σ_0^2 is a known constant, the term $\sum x_i^2$ in the exponent does not involve any unknown parameter. We can absorb it into $h(\mathbf{x})$:

$$f(\mathbf{x}|\mu) = \underbrace{\exp\left(\frac{2\mu\sum x_i - n\mu^2}{2\sigma_0^2}\right)}_{g(\sum x_i,\, \mu)} \cdot \underbrace{\frac{1}{(2\pi\sigma_0^2)^{n/2}}\exp\left(-\frac{\sum x_i^2}{2\sigma_0^2}\right)}_{h(\mathbf{x})}.$$

By the factorization theorem, $T = \sum X_i$ (equivalently $\bar{X}$) is sufficient for μ.

3. $X_1, \ldots, X_n \sim$ iid Gamma(α, β), both unknown.

   Solution. Each observation has pdf $f(x_i|\alpha,\beta) = \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} e^{-x_i/\beta}$ for $x_i > 0$. The joint density is:

$$f(\mathbf{x}|\alpha,\beta) = \prod_{i=1}^{n} \frac{1}{\Gamma(\alpha)\beta^\alpha} x_i^{\alpha-1} e^{-x_i/\beta} = \frac{1}{\Gamma(\alpha)^n \beta^{n\alpha}} \left(\prod_{i=1}^{n} x_i\right)^{\alpha-1} \exp\left(-\frac{\sum x_i}{\beta}\right).$$

   Rewrite the product $\prod x_i^{\alpha-1} = \exp((\alpha-1)\sum \log x_i)$ to make the dependence on α clearer:

$$f(\mathbf{x}|\alpha,\beta) = \underbrace{\frac{1}{\Gamma(\alpha)^n\beta^{n\alpha}}\exp\left((\alpha-1)\sum_{i=1}^{n}\log x_i - \frac{\sum_{i=1}^{n} x_i}{\beta}\right)}_{g\left(\sum x_i,\, \sum \log x_i,\, \alpha,\, \beta\right)} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

   The joint density depends on the data only through $\sum x_i$ and $\sum \log x_i$, so $T = (\sum X_i, \sum \log X_i)$ is sufficient for (α, β).

4. $X_1, \ldots, X_n \sim$ iid Uniform(α, β), both unknown.

   Solution. Each observation has pdf $f(x_i|\alpha,\beta) = \frac{1}{\beta-\alpha}\mathbf{1}_{(\alpha,\beta)}(x_i)$. The joint density is:

$$f(\mathbf{x}|\alpha,\beta) = \frac{1}{(\beta-\alpha)^n}\prod_{i=1}^{n}\mathbf{1}_{(\alpha,\beta)}(x_i).$$

   The product of indicators requires every x_i to lie in (α, β). This holds if and only if the smallest observation exceeds α and the largest is below β:

$$\prod_{i=1}^{n}\mathbf{1}_{(\alpha,\beta)}(x_i) = \mathbf{1}_{(\alpha<x_{(1)})}\cdot\mathbf{1}_{(x_{(n)}<\beta)}.$$

So we can factor:

$$f(\mathbf{x}|\alpha,\beta) = \underbrace{\frac{1}{(\beta-\alpha)^n}\mathbf{1}_{(\alpha<x_{(1)})}\mathbf{1}_{(x_{(n)}<\beta)}}_{g(x_{(1)},x_{(n)},\alpha,\beta)} \cdot \underbrace{1}_{h(\mathbf{x})}.$$

By the factorization theorem, $T = (X_{(1)}, X_{(n)})$ is sufficient for (α, β).

Drill: Rao–Blackwell in Action

1. Let $X_1, \ldots, X_n \sim$ iid Bernoulli(p). Start with the crude estimator $W = X_1$. The sufficient statistic is $T = \sum X_i$. Compute $\phi(T) = \mathbb{E}[X_1 \mid \sum X_i = t]$ and verify it equals $t/n = \bar{X}$.

   Solution. We need $\mathbb{P}(X_1 = 1 \mid \sum_{i=1}^n X_i = t)$. By the definition of conditional probability:

   $$\mathbb{P}(X_1 = 1 \mid T = t) = \frac{\mathbb{P}(X_1 = 1, T = t)}{\mathbb{P}(T = t)}.$$

   The event $\{X_1 = 1, T = t\}$ means $X_1 = 1$ and the remaining $n-1$ observations sum to $t-1$. Since the X_i are independent:

   $$\mathbb{P}(X_1 = 1, T = t) = p \cdot \binom{n-1}{t-1} p^{t-1}(1-p)^{n-t} = \binom{n-1}{t-1} p^t (1-p)^{n-t}.$$

   The denominator is:

   $$\mathbb{P}(T = t) = \binom{n}{t} p^t (1-p)^{n-t}.$$

   Dividing:

   $$\mathbb{P}(X_1 = 1 \mid T = t) = \frac{\binom{n-1}{t-1}}{\binom{n}{t}} = \frac{(n-1)!\,/\,((t-1)!(n-t)!)}{n!\,/\,(t!(n-t)!)} = \frac{t}{n}.$$

   So $\phi(T) = \mathbb{E}[X_1 \mid T] = T/n = \bar{X}$.

   Symmetry shortcut. Since $X_1, \ldots, X_n$ are iid, $\mathbb{E}[X_i \mid T]$ is the same for every i. Summing over i:

   $$\sum_{i=1}^n \mathbb{E}[X_i \mid T] = \mathbb{E}[T \mid T] = T.$$

   By symmetry each term equals T/n, so $\mathbb{E}[X_1 \mid T] = T/n = \bar{X}$.

   Since $T = \sum X_i$ is complete sufficient for Bernoulli(p), the Rao–Blackwellized estimator $\bar{X}$ is the UMVUE of p.

2. Let $X_1, \ldots, X_n \sim$ iid Exp(λ). $T = \sum X_i$ is complete sufficient. We know $2\lambda \sum X_i \sim \chi^2_{2n}$. The estimator $W = \mathbf{1}_{\{X_1 > a\}}$ is unbiased for $e^{-\lambda a}$. Compute $\mathbb{E}[W \mid T]$ to find the UMVUE of $e^{-\lambda a}$.

   Solution. We need to compute $\mathbb{E}\Big[\mathbf{1}_{\{X_1>a\}} \mid T = t\Big]$ where $T = \sum_{i=1}^n X_i = t$.

   Define $S = \sum_{i=2}^n X_i = T - X_1$. Since X_1 is independent of $X_2, \ldots, X_n$, we have S independent of X_1.

If $T = t \le a$, then $X_1 \le t \le a$ (since $X_1 \le T$ when all observations are nonnegative), so $\mathbf{1}_{\{X_1 > a\}} = 0$ and $\mathbb{E}[W \mid T = t] = 0$.

Now suppose $t > a$. We compute $\mathbb{P}(X_1 > a \mid X_1 + S = t)$. The conditional density of X_1 given $X_1 + S = t$ is the ratio of joint to marginal densities.

The key fact is: $X_1 \sim \text{Exp}(\lambda)$ and $S = \sum_{i=2}^{n} X_i \sim \text{Gamma}(n-1, 1/\lambda)$ are independent, and $T = X_1 + S \sim \text{Gamma}(n, 1/\lambda)$.

The conditional density of X_1 given $T = t$ is:

$$f_{X_1|T}(x \mid t) = \frac{f_{X_1}(x) \cdot f_S(t-x)}{f_T(t)}, \quad 0 < x < t.$$

Substituting the densities:

$$f_{X_1}(x) = \lambda e^{-\lambda x}, \qquad f_S(s) = \frac{\lambda^{n-1} s^{n-2} e^{-\lambda s}}{\Gamma(n-1)}, \qquad f_T(t) = \frac{\lambda^n t^{n-1} e^{-\lambda t}}{\Gamma(n)}.$$

$$f_{X_1|T}(x \mid t) = \frac{\lambda e^{-\lambda x} \cdot \frac{\lambda^{n-1}(t-x)^{n-2} e^{-\lambda(t-x)}}{\Gamma(n-1)}}{\frac{\lambda^n t^{n-1} e^{-\lambda t}}{\Gamma(n)}}.$$

The exponential terms cancel: $e^{-\lambda x} e^{-\lambda(t-x)} = e^{-\lambda t}$, and λ^n cancels in numerator and denominator. Using $\Gamma(n) = (n-1)\Gamma(n-1)$:

$$f_{X_1|T}(x \mid t) = \frac{(n-1)(t-x)^{n-2}}{t^{n-1}}, \quad 0 < x < t.$$

This is a Beta-type density on $(0, t)$: given $T = t$, the conditional distribution of X_1 has density $(n-1)(1 - x/t)^{n-2}/t$.

Now compute the conditional probability:

$$\mathbb{P}(X_1 > a \mid T = t) = \int_a^t \frac{(n-1)(t-x)^{n-2}}{t^{n-1}}\, dx.$$

Substitute $u = t - x$, $du = -dx$. When $x = a$, $u = t - a$; when $x = t$, $u = 0$:

$$= \frac{n-1}{t^{n-1}} \int_0^{t-a} u^{n-2}\, du = \frac{n-1}{t^{n-1}} \cdot \frac{(t-a)^{n-1}}{n-1} = \left(\frac{t-a}{t}\right)^{n-1} = \left(1 - \frac{a}{t}\right)^{n-1}.$$

Combining both cases, the Rao–Blackwellized estimator is:

$$\mathbb{E}[W \mid T] = \left(1 - \frac{a}{T}\right)^{n-1} \mathbf{1}_{\{T > a\}}.$$

Since $T = \sum X_i$ is complete sufficient, this is the UMVUE of $e^{-\lambda a}$.

Drill: Cramér–Rao Bound

1. Verify that $\bar{X}$ attains the Cramér–Rao bound for Bernoulli(p): compute $I(p) = 1/(p(1-p))$ and confirm $\mathrm{Var}\left(\bar{X}\right) = p(1-p)/n = 1/(nI(p))$.

   Solution. The pmf of a single Bernoulli observation is $f(x|p) = p^x(1-p)^{1-x}$ for $x \in \{0, 1\}$. The log-likelihood for one observation is:

   $$\log f(x|p) = x \log p + (1-x)\log(1-p).$$

   The score function is:

   $$\frac{\partial}{\partial p} \log f(x|p) = \frac{x}{p} - \frac{1-x}{1-p} = \frac{x-p}{p(1-p)}.$$

   The Fisher information is the variance of the score (equivalently, the expected squared score):

   $$I(p) = \mathbb{E}\left[\left(\frac{X-p}{p(1-p)}\right)^2\right] = \frac{\mathrm{Var}(X)}{(p(1-p))^2} = \frac{p(1-p)}{(p(1-p))^2} = \frac{1}{p(1-p)}.$$

   Alternatively, using the second derivative:

   $$\frac{\partial^2}{\partial p^2} \log f(x|p) = -\frac{x}{p^2} - \frac{1-x}{(1-p)^2}.$$

   $$I(p) = -\mathbb{E}\left[\frac{\partial^2}{\partial p^2} \log f(X|p)\right] = \frac{p}{p^2} + \frac{1-p}{(1-p)^2} = \frac{1}{p} + \frac{1}{1-p} = \frac{1}{p(1-p)}.$$

   The Cramér–Rao lower bound for an unbiased estimator of p based on n iid observations is:

   $$\mathrm{Var}(W) \geq \frac{1}{nI(p)} = \frac{p(1-p)}{n}.$$

   Since $\bar{X}$ is unbiased for p (as $\mathbb{E}\left[\bar{X}\right] = p$) and:

   $$\mathrm{Var}\left(\bar{X}\right) = \frac{\mathrm{Var}(X_1)}{n} = \frac{p(1-p)}{n} = \frac{1}{nI(p)},$$

   the sample mean $\bar{X}$ attains the Cramér–Rao bound exactly, confirming it is the best unbiased estimator of p.

2. For Exp(λ) with rate parametrization: compute $I(\lambda) = 1/\lambda^2$. What is the CRLB for unbiased estimation of λ? Does $1/\bar{X}$ attain it?

   Solution. The pdf is $f(x|\lambda) = \lambda e^{-\lambda x}$ for $x > 0$. The log-likelihood for one observation is:

   $$\log f(x|\lambda) = \log \lambda - \lambda x.$$

   Using the second-derivative method:

   $$\frac{d^2}{d\lambda^2} \log f(x|\lambda) = -\frac{1}{\lambda^2}.$$

This is a constant, so:

$$I(\lambda) = -\mathbb{E}\left[-\frac{1}{\lambda^2}\right] = \frac{1}{\lambda^2}.$$

The Cramér–Rao lower bound for any unbiased estimator of λ based on n iid observations is:

$$\mathrm{Var}(W) \geq \frac{1}{nI(\lambda)} = \frac{\lambda^2}{n}.$$

Now consider $1/\bar{X}$. The MLE is $\hat{\lambda} = 1/\bar{X}$, but it is **biased**. To see this: $\bar{X} = \frac{1}{n}\sum X_i$ where $\sum X_i \sim \mathrm{Gamma}(n, 1/\lambda)$. By Jensen's inequality, $\mathbb{E}\left[1/\bar{X}\right] > 1/\mathbb{E}\left[\bar{X}\right] = \lambda$ (since $1/x$ is convex for $x > 0$).

More precisely, $2\lambda \sum X_i \sim \chi^2_{2n}$, so $\bar{X} = \frac{1}{2n\lambda} \cdot (2\lambda \sum X_i)$ and:

$$\mathbb{E}\left[\frac{1}{\bar{X}}\right] = 2n\lambda\, \mathbb{E}\left[\frac{1}{\chi^2_{2n}}\right] \cdot \frac{1}{2n} = \lambda \cdot \frac{2n}{2n-2} = \frac{n\lambda}{n-1}.$$

Since $1/\bar{X}$ is biased, the standard CRLB comparison does not directly apply. The unbiased version $\frac{n-1}{n} \cdot \frac{1}{\bar{X}}$ is the natural candidate, but its variance exceeds λ^2/n, so neither version attains the bound. In fact, no unbiased estimator of λ attains the CRLB in the exponential rate family.

3. For $N(\mu, \sigma^2)$ with σ^2 known: compute $I(\mu) = 1/\sigma^2$ and confirm $\mathrm{Var}\left(\bar{X}\right) = \sigma^2/n = 1/(nI(\mu))$.

   Solution. The pdf of a single observation is $f(x|\mu) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$. The log-likelihood for one observation is:

$$\log f(x|\mu) = -\frac{1}{2}\log(2\pi\sigma^2) - \frac{(x-\mu)^2}{2\sigma^2}.$$

   The score function is:

$$\frac{\partial}{\partial\mu} \log f(x|\mu) = \frac{x-\mu}{\sigma^2}.$$

   Using the second-derivative formula for Fisher information:

$$\frac{\partial^2}{\partial\mu^2} \log f(x|\mu) = -\frac{1}{\sigma^2}.$$

   This is a constant (does not depend on x), so:

$$I(\mu) = -\mathbb{E}\left[\frac{\partial^2}{\partial\mu^2} \log f(X|\mu)\right] = \frac{1}{\sigma^2}.$$

   The Cramér–Rao lower bound for unbiased estimation of μ from n iid observations is:

$$\mathrm{Var}(W) \geq \frac{1}{nI(\mu)} = \frac{\sigma^2}{n}.$$

Since $\bar{X}$ is unbiased for μ and:

$$\mathrm{Var}\left(\bar{X}\right) = \frac{\sigma^2}{n} = \frac{1}{nI(\mu)},$$

the sample mean $\bar{X}$ attains the Cramér–Rao bound, confirming it is the best unbiased estimator of μ when σ^2 is known.

6 Hypothesis Testing — Problems

6.1 Likelihood Ratio Tests

Example: Normal LRT — Known Variance

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\theta, \sigma^2)$ with σ^2 known. Test $H_0\colon \theta = \theta_0$ versus $H_1\colon \theta \neq \theta_0$.

Step 1: Unrestricted MLE. The MLE over $\Theta = \mathbb{R}$ is $\hat{\theta} = \bar{X}$.
Step 2: Restricted MLE. Under H_0, the only value is θ_0, so $\hat{\theta}_0 = \theta_0$.
Step 3: Form the ratio.

$$\lambda(\mathbf{x}) = \frac{L(\theta_0 \mid \mathbf{x})}{L(\bar{x} \mid \mathbf{x})} = \frac{(2\pi\sigma^2)^{-n/2} \exp[-\frac{1}{2\sigma^2}\sum(x_i - \theta_0)^2]}{(2\pi\sigma^2)^{-n/2} \exp[-\frac{1}{2\sigma^2}\sum(x_i - \bar{x})^2]}.$$

The $(2\pi\sigma^2)^{-n/2}$ terms cancel. Using the identity $\sum(x_i - \theta_0)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - \theta_0)^2$:

$$\lambda(\mathbf{x}) = \exp\Big[-\frac{n(\bar{x} - \theta_0)^2}{2\sigma^2}\Big].$$

Step 4: Simplify the rejection region.
Reject when $\lambda(\mathbf{x}) \leq c$. Taking logs:

$$-\frac{n(\bar{x} - \theta_0)^2}{2\sigma^2} \leq \log c \quad \Longleftrightarrow \quad |\bar{x} - \theta_0| \geq \sqrt{-2\sigma^2(\log c)/n}.$$

Define the z-statistic $Z = \dfrac{\bar{X} - \theta_0}{\sigma/\sqrt{n}}$. The rejection region is $|Z| \geq z_{\alpha/2}$.
Step 5: Choose the cutoff.
Under H_0, $Z \sim \mathrm{n}(0,1)$, so $\mathbb{P}\Big(|Z| \geq z_{\alpha/2}\Big) = \alpha$. The test has exact size α.

Example: Normal LRT — Unknown Variance (the t-test)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$ with *both* μ and σ^2 unknown. Test $H_0\colon \mu \leq \mu_0$ versus $H_1\colon \mu > \mu_0$. Here σ^2 is a **nuisance parameter**.

Unrestricted MLEs: $\hat{\mu} = \bar{X}$ and $\hat{\sigma}^2 = \frac{1}{n}\sum(X_i - \bar{X})^2$.
Restricted MLEs (maximize over $\mu \leq \mu_0, \sigma^2 > 0$):

- If $\bar{X} \leq \mu_0$: the unrestricted MLE already satisfies the constraint, so $\hat{\mu}_0 = \bar{X}$ and $\lambda(\mathbf{x}) = 1$ (do not reject).
- If $\bar{X} > \mu_0$: the restricted MLE is $\hat{\mu}_0 = \mu_0$ with $\hat{\sigma}_0^2 = \frac{1}{n}\sum(x_i - \mu_0)^2$.

After algebra, one can show that the rejection region $\lambda(\mathbf{x}) \leq c$ is equivalent to

$$T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}} \;\geq\; t_{n-1,\alpha},$$

where $S^2 = \frac{1}{n-1}\sum(X_i - \bar{X})^2$ and $T \sim t_{n-1}$ under $H_0\colon \mu = \mu_0$. This is the familiar **one-sided one-sample t-test**.

6.2 The Neyman–Pearson Lemma

Example: Neyman–Pearson for Bernoulli (The Coin Flip)

Let $X \sim \text{Binomial}(5, \theta)$. Test $H_0 : \theta = \frac{1}{2}$ versus $H_1 : \theta = \frac{3}{4}$.

Step 1: Compute the likelihood ratio for each x.
The pmf is $f(x|\theta) = \binom{5}{x}\theta^x(1-\theta)^{5-x}$. The NP Lemma says: reject when $f(x|\theta_1)/f(x|\theta_0) > k$, i.e. when

$$\frac{f(x \mid \frac{3}{4})}{f(x \mid \frac{1}{2})} = \frac{(\frac{3}{4})^x(\frac{1}{4})^{5-x}}{(\frac{1}{2})^5} = \frac{3^x}{4^5} \cdot 2^5 = \frac{3^x}{32} > k.$$

This ratio is *increasing in* x, so the NP test rejects for large x.

Step 2: Choose the cutoff for size α.

x	0	1	2	3	4	5
$f(x\|\frac{3}{4})/f(x\|\frac{1}{2})$	1/32	3/32	9/32	27/32	81/32	243/32

If we want $\alpha = P_{1/2}(X \geq 5) = (1/2)^5 = 1/32 \approx 0.031$: reject if $X = 5$.
If we want $\alpha = P_{1/2}(X \geq 4) = 6/32 \approx 0.188$: reject if $X \geq 4$.

Power at $\theta_1 = 3/4$:

$$\beta(3/4) = P_{3/4}(X \geq 4) = \binom{5}{4}\left(\tfrac{3}{4}\right)^4\left(\tfrac{1}{4}\right) + \left(\tfrac{3}{4}\right)^5 = \frac{405}{1024} + \frac{243}{1024} = \frac{648}{1024} \approx 0.633.$$

No other level 6/32 test can have power greater than 0.633 at $\theta = 3/4$.

6.3 Power Functions

Example: How Power Changes with n

Fix $\theta_0 = 0$, $\sigma = 1$, $\alpha = 0.05$ (so $c = z_{0.05} = 1.645$). The power at $\theta = 0.5$ is:

$$\beta(0.5) = 1 - \Phi\left(1.645 - \frac{0.5}{1/\sqrt{n}}\right) = 1 - \Phi(1.645 - 0.5\sqrt{n}).$$

n	Argument of Φ	$\beta(0.5)$
1	1.145	0.126
10	0.064	0.474
25	−0.855	0.804
50	−1.891	0.971
100	−3.355	0.9996

Takeaway: Power increases with n. To detect a shift of $\delta = \theta - \theta_0$, you need $n \approx (z_\alpha + z_\beta)^2\sigma^2/\delta^2$, where $1 - \beta$ is the desired power.

Example: Power of the Two-Sided z-test

Test $H_0\colon \theta = \theta_0$ vs. $H_1\colon \theta \neq \theta_0$. Reject if $|Z| \geq z_{\alpha/2}$. The power function is:

$$\beta(\theta) = \Phi\left(-z_{\alpha/2} + \frac{\theta - \theta_0}{\sigma/\sqrt{n}}\right) + 1 - \Phi\left(z_{\alpha/2} + \frac{\theta - \theta_0}{\sigma/\sqrt{n}}\right).$$

This is minimized at $\theta = \theta_0$ (where $\beta(\theta_0) = \alpha$) and increases symmetrically as $|\theta - \theta_0|$ grows.

6.4 Monotone Likelihood Ratio and Karlin–Rubin

Example: UMP Test for Normal Mean

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\theta, \sigma^2)$ with σ^2 known. $\bar{X}$ is sufficient, and the normal family has MLR in $\bar{x}$. By Karlin–Rubin:

- $H_0\colon \theta \leq \theta_0$ vs. $H_1\colon \theta > \theta_0$: UMP rejects when $\bar{X} > \theta_0 + z_\alpha\, \sigma/\sqrt{n}$.
- $H_0\colon \theta \geq \theta_0$ vs. $H_1\colon \theta < \theta_0$: UMP rejects when $\bar{X} < \theta_0 - z_\alpha\, \sigma/\sqrt{n}$.

For the **two-sided** test $H_0\colon \theta = \theta_0$ vs. $H_1\colon \theta \neq \theta_0$: no UMP test exists!

6.5 p-Values

Example: Normal p-Values

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$.

- **One-sided** ($H_0\colon \mu \leq \mu_0$ vs. $H_1\colon \mu > \mu_0$):

$$p = P(T_{n-1} \geq t_{\text{obs}}) = 1 - F_{t_{n-1}}(t_{\text{obs}}), \quad t_{\text{obs}} = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}.$$

- **Two-sided** ($H_0\colon \mu = \mu_0$ vs. $H_1\colon \mu \neq \mu_0$):

$$p = 2\, P(T_{n-1} \geq |t_{\text{obs}}|).$$

6.6 Sample Size Determination

Example: Choosing n for a Clinical Trial

Want to detect a shift of $\delta = 0.5$ (in units of $\sigma = 1$) with $\alpha = 0.05$ and power 0.80.
We need $z_{0.05} = 1.645$ and $z_{0.20} = 0.842$.

$$n = \frac{(1.645 + 0.842)^2 \cdot 1}{0.5^2} = \frac{(2.487)^2}{0.25} = \frac{6.185}{0.25} \approx 24.7.$$

Round up: $n = 25$.

6.7 Drill Problems

Drill: Construct an LRT from Scratch

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$ with σ^2 known.

(a) Test $H_0\colon \mu = 3$ vs. $H_1\colon \mu \neq 3$. Derive $\lambda(\mathbf{x})$ and show the rejection region is $|\bar{x} - 3| > z_{\alpha/2}\,\sigma/\sqrt{n}$.

(b) Compute the power at $\mu = 4$ when $\sigma = 2$, $n = 16$, $\alpha = 0.05$.

Solution:
(a) The likelihood under the full parameter space is maximized at $\hat{\mu} = \bar{x}$, while under H_0 the only allowed value is $\mu = 3$. The LRT statistic is therefore

$$\lambda(\mathbf{x}) = \frac{L(3 \mid \mathbf{x})}{L(\bar{x} \mid \mathbf{x})} = \frac{\exp[-\frac{1}{2\sigma^2}\sum(x_i - 3)^2]}{\exp[-\frac{1}{2\sigma^2}\sum(x_i - \bar{x})^2]}.$$

Using the identity $\sum(x_i - 3)^2 = \sum(x_i - \bar{x})^2 + n(\bar{x} - 3)^2$, the ratio simplifies to

$$\lambda(\mathbf{x}) = \exp\Big[-\frac{n(\bar{x} - 3)^2}{2\sigma^2}\Big].$$

We reject when $\lambda(\mathbf{x}) \leq c$. Taking logarithms:

$$-\frac{n(\bar{x} - 3)^2}{2\sigma^2} \leq \log c \quad \Longleftrightarrow \quad (\bar{x} - 3)^2 \geq -\frac{2\sigma^2 \log c}{n}.$$

This is equivalent to $|\bar{x} - 3| \geq c'$ for some constant $c' > 0$. Define $Z = \frac{\bar{X} - 3}{\sigma/\sqrt{n}}$, so the rejection region becomes $|Z| \geq z_{\alpha/2}$. Under H_0, $Z \sim \mathrm{n}(0, 1)$, which gives exact size α. Equivalently, we reject when $|\bar{x} - 3| > z_{\alpha/2}\,\sigma/\sqrt{n}$.
(b) With $\sigma = 2$, $n = 16$, and $\alpha = 0.05$ we have $z_{\alpha/2} = 1.96$ and $\sigma/\sqrt{n} = 2/4 = 0.5$. Under $\mu = 4$, the test statistic is $Z = \frac{\bar{X} - 3}{0.5} \sim \mathrm{n}(2, 1)$. The power is

$$\beta(4) = \mathbb{P}(|Z| \geq 1.96) = \mathbb{P}(Z \geq 1.96) + \mathbb{P}(Z \leq -1.96)\,.$$

Since $Z \sim \mathrm{n}(2, 1)$:

$$\mathbb{P}(Z \geq 1.96) = 1 - \Phi(1.96 - 2) = 1 - \Phi(-0.04) \approx 0.516.$$

For the left tail:

$$\mathbb{P}(Z \leq -1.96) = \Phi(-1.96 - 2) = \Phi(-3.96) \approx 0.00004.$$

Therefore $\beta(4) \approx 0.516$. (The left tail contributes negligibly.)

Drill: Neyman–Pearson Mechanics

Let $X \sim \mathrm{Poisson}(\lambda)$. Test $H_0\colon \lambda = 1$ vs. $H_1\colon \lambda = 3$.

(a) Write down the likelihood ratio $f(x|3)/f(x|1)$ and show it is increasing in x.

(b) Find the rejection region for size $\alpha \le 0.08$. (*Hint:* $P(X \ge 3 \mid \lambda = 1) = 1 - e^{-1}(1+1+1/2) \approx 0.080$.)

(c) Compute the power of this test.

Solution:
(a) The Poisson pmf is $f(x|\lambda) = e^{-\lambda}\lambda^x/x!$. The likelihood ratio is

$$\frac{f(x|3)}{f(x|1)} = \frac{e^{-3}\,3^x/x!}{e^{-1}\,1^x/x!} = e^{-2}\,3^x.$$

Since 3^x is increasing in x, the ratio $e^{-2}\,3^x$ is increasing in x. By the Neyman–Pearson Lemma, the most powerful test rejects for large values of x.
(b) We need the largest rejection region $\{X \ge c\}$ such that $P_{\lambda=1}(X \ge c) \le 0.08$. Under $\lambda = 1$:

$$P(X \ge 3 \mid \lambda = 1) = 1 - P(X \le 2 \mid \lambda = 1) = 1 - e^{-1}\Big(1 + 1 + \tfrac{1}{2}\Big) = 1 - \tfrac{5}{2}e^{-1} \approx 1 - 0.920 = 0.080.$$

So $\alpha = P_1(X \ge 3) \approx 0.080 \le 0.08$, and the rejection region is $\{X \ge 3\}$. (Using $\{X \ge 2\}$ would give $P_1(X \ge 2) = 1 - 2e^{-1} \approx 0.264$, which exceeds 0.08.)
(c) The power is $P_{\lambda=3}(X \ge 3)$. Under $\lambda = 3$:

$$P(X \le 2 \mid \lambda = 3) = e^{-3}\Big(1 + 3 + \tfrac{9}{2}\Big) = e^{-3} \cdot \tfrac{17}{2} \approx 0.0498 \cdot 8.5 \approx 0.423.$$

Therefore the power is

$$\beta(3) = 1 - 0.423 = 0.577.$$

Drill: Power Function Computation

$X_1, \ldots, X_{25} \overset{\text{iid}}{\sim} \mathrm{n}(\theta, 4)$. Test $H_0\colon \theta \le 0$ vs. $H_1\colon \theta > 0$ at level $\alpha = 0.05$.

(a) Write down the rejection region.

(b) Compute $\beta(\theta)$ as a function of θ.

(c) Evaluate $\beta(0)$, $\beta(0.5)$, $\beta(1)$.

(d) What sample size is needed for $\beta(1) \ge 0.90$?

Solution:
We have $n = 25$, $\sigma^2 = 4$ (so $\sigma = 2$), $\alpha = 0.05$, and $z_{0.05} = 1.645$.
(a) The test statistic is $Z = \frac{\bar{X} - 0}{\sigma/\sqrt{n}} = \frac{\bar{X}}{2/5} = \frac{5\bar{X}}{2}$. Under $H_0\colon \theta = 0$, $Z \sim \mathrm{n}(0, 1)$. The rejection region is

$$Z \ge z_{0.05} = 1.645 \quad \Longleftrightarrow \quad \bar{X} \ge 1.645 \cdot \frac{2}{5} = 0.658.$$

(b) Under θ, $\bar{X} \sim \mathrm{n}(\theta, 4/25)$, so $Z = \frac{\bar{X}}{2/5} \sim \mathrm{n}\Big(\frac{5\theta}{2}, 1\Big)$. The power function is

$$\beta(\theta) = \mathbb{P}(Z \ge 1.645) = 1 - \Phi\Big(1.645 - \frac{5\theta}{2}\Big).$$

(c) Evaluating at the requested values:
At $\theta = 0$:

$$\beta(0) = 1 - \Phi(1.645) = 1 - 0.95 = 0.05.$$

At $\theta = 0.5$:

$$\beta(0.5) = 1 - \Phi(1.645 - 1.25) = 1 - \Phi(0.395) \approx 1 - 0.654 = 0.346.$$

At $\theta = 1$:

$$\beta(1) = 1 - \Phi(1.645 - 2.5) = 1 - \Phi(-0.855) \approx 1 - 0.196 = 0.804.$$

(d) We need $\beta(1) \geq 0.90$, i.e.,

$$1 - \Phi\left(1.645 - \frac{\sqrt{n}}{2}\right) \geq 0.90.$$

This requires $\Phi\left(1.645 - \frac{\sqrt{n}}{2}\right) \leq 0.10$, so

$$1.645 - \frac{\sqrt{n}}{2} \leq -1.282.$$

Solving for n:

$$\frac{\sqrt{n}}{2} \geq 1.645 + 1.282 = 2.927 \quad \Longrightarrow \quad \sqrt{n} \geq 5.854 \quad \Longrightarrow \quad n \geq 34.3.$$

Rounding up, $n = 35$.

Drill: Identify the UMP Test via Karlin–Rubin

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exponential}(\theta)$ where $f(x|\theta) = \frac{1}{\theta} e^{-x/\theta}$, $x > 0$.

(a) Show that $T = \sum X_i$ is sufficient and the family has MLR in T (with T increasing as θ increases).

(b) State the UMP level α test of $H_0\colon \theta \leq \theta_0$ vs. $H_1\colon \theta > \theta_0$. (*Hint:* $2\sum X_i/\theta_0 \sim \chi^2_{2n}$ under $\theta = \theta_0$.)

(c) Why is there no UMP test of $H_0\colon \theta = \theta_0$ vs. $H_1\colon \theta \neq \theta_0$?

Solution:
(a) The joint density is

$$f(\mathbf{x}|\theta) = \prod_{i=1}^{n} \frac{1}{\theta} e^{-x_i/\theta} = \frac{1}{\theta^n} \exp\Big(-\frac{\sum x_i}{\theta}\Big).$$

By the factorization theorem, $T = \sum X_i$ is sufficient for θ. This is an exponential family with natural parameter $\eta(\theta) = -1/\theta$ and sufficient statistic T. Since $\eta(\theta) = -1/\theta$ is strictly increasing in θ (for $\theta > 0$), the family has monotone likelihood ratio in T.
To verify directly: for $\theta_1 > \theta_0 > 0$, the likelihood ratio is

$$\frac{f(\mathbf{x}|\theta_1)}{f(\mathbf{x}|\theta_0)} = \left(\frac{\theta_0}{\theta_1}\right)^n \exp\Big[-\Big(\frac{1}{\theta_1} - \frac{1}{\theta_0}\Big) T\Big] = \left(\frac{\theta_0}{\theta_1}\right)^n \exp\Big[\Big(\frac{1}{\theta_0} - \frac{1}{\theta_1}\Big) T\Big].$$

Since $\theta_1 > \theta_0 > 0$, we have $1/\theta_0 > 1/\theta_1$, so the coefficient $1/\theta_0 - 1/\theta_1 > 0$. The exponent is a positive constant times T, so the ratio is increasing in T. This confirms MLR in T.
(b) By the Karlin–Rubin theorem, the UMP level α test of $H_0\colon \theta \le \theta_0$ vs. $H_1\colon \theta > \theta_0$ rejects for large values of $T = \sum X_i$.
Under $\theta = \theta_0$, each $X_i \sim \text{Exponential}(\theta_0)$, so $X_i/\theta_0 \sim \text{Exponential}(1)$ and $2X_i/\theta_0 \sim \chi^2_2$. By independence:

$$\frac{2T}{\theta_0} = \frac{2\sum X_i}{\theta_0} \sim \chi^2_{2n}.$$

The UMP test therefore rejects when

$$\frac{2\sum X_i}{\theta_0} \ge \chi^2_{2n,\alpha},$$

where $\chi^2_{2n,\alpha}$ is the upper α quantile of the χ^2_{2n} distribution.
(c) For the two-sided alternative $H_1\colon \theta \ne \theta_0$, no UMP test exists. The Karlin–Rubin theorem only applies to one-sided alternatives. Against $\theta > \theta_0$ the UMP test rejects for large T, but against $\theta < \theta_0$ the UMP test rejects for small T. These two rejection regions are incompatible: any single test that is most powerful against all alternatives on one side cannot simultaneously be most powerful against all alternatives on the other side.

Drill: LRT with Nuisance Parameters

$X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{n}(\mu, \sigma^2)$, both unknown. Test $H_0\colon \mu = \mu_0$ vs. $H_1\colon \mu \ne \mu_0$.

(a) Find the unrestricted MLEs $(\hat\mu, \hat\sigma^2)$.

(b) Find the restricted MLEs under H_0.

(c) Derive the LRT statistic $\lambda(\mathbf{x})$ and show the rejection region reduces to $|(\bar x - \mu_0)/(s/\sqrt{n})| > t_{n-1,\alpha/2}$.

Solution:
(a) The full parameter space is $\Theta = \{(\mu, \sigma^2) : \mu \in \mathbb{R},\ \sigma^2 > 0\}$. Maximizing the normal likelihood over Θ gives the unrestricted MLEs:

$$\hat\mu = \bar X, \qquad \hat\sigma^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \bar X)^2.$$

(b) Under $H_0\colon \mu = \mu_0$, we fix $\mu = \mu_0$ and maximize over $\sigma^2 > 0$ only. The restricted MLE of σ^2 is

$$\hat\sigma_0^2 = \frac{1}{n}\sum_{i=1}^n (X_i - \mu_0)^2.$$

(c) The LRT statistic is

$$\lambda(\mathbf{x}) = \frac{L(\mu_0, \hat\sigma_0^2)}{L(\hat\mu, \hat\sigma^2)} = \frac{(2\pi\hat\sigma_0^2)^{-n/2}\exp(-n/2)}{(2\pi\hat\sigma^2)^{-n/2}\exp(-n/2)} = \left(\frac{\hat\sigma^2}{\hat\sigma_0^2}\right)^{n/2}.$$

Now expand $\hat\sigma_0^2$ using the identity $\sum(x_i - \mu_0)^2 = \sum(x_i - \bar x)^2 + n(\bar x - \mu_0)^2$:

$$\hat\sigma_0^2 = \frac{1}{n}\left[n\hat\sigma^2 + n(\bar x - \mu_0)^2\right] = \hat\sigma^2 + (\bar x - \mu_0)^2.$$

Substituting:

$$\lambda(\mathbf{x}) = \left(\frac{\hat{\sigma}^2}{\hat{\sigma}^2 + (\bar{x} - \mu_0)^2}\right)^{n/2} = \left(\frac{1}{1 + (\bar{x} - \mu_0)^2/\hat{\sigma}^2}\right)^{n/2}.$$

Since $\hat{\sigma}^2 = \frac{n-1}{n} s^2$ where $s^2 = \frac{1}{n-1}\sum(x_i - \bar{x})^2$, we have

$$\frac{(\bar{x} - \mu_0)^2}{\hat{\sigma}^2} = \frac{n}{n-1} \cdot \frac{(\bar{x} - \mu_0)^2}{s^2} = \frac{1}{n-1} \cdot \frac{(\bar{x} - \mu_0)^2}{s^2/n} = \frac{t^2}{n-1},$$

where $t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}}$. Therefore

$$\lambda(\mathbf{x}) = \left(1 + \frac{t^2}{n-1}\right)^{-n/2}.$$

This is a decreasing function of $|t|$, so $\lambda(\mathbf{x}) \le c$ is equivalent to $|t| \ge c'$ for some cutoff c'. Under $H_0\colon \mu = \mu_0$, the statistic $T = \frac{\bar{X} - \mu_0}{S/\sqrt{n}}$ follows a t_{n-1} distribution. Setting $c' = t_{n-1,\alpha/2}$ gives exact size α, and the rejection region is

$$\left|\frac{\bar{x} - \mu_0}{s/\sqrt{n}}\right| > t_{n-1,\alpha/2}.$$

This is the two-sided one-sample t-test.

Drill: Discrete Neyman–Pearson

$X \sim \text{Binomial}(4, \theta)$. Test $H_0\colon \theta = 0.5$ vs. $H_1\colon \theta = 0.8$.

(a) Compute $f(x|0.8)/f(x|0.5)$ for $x = 0, 1, 2, 3, 4$.

(b) Find the UMP test of size $\alpha = P_{0.5}(X \ge 4) = 1/16$.

(c) Compute the power of this test at $\theta = 0.8$.

(d) What is the UMP test of size $\alpha = 5/16$? (*Hint:* you may need a randomized test.)

Solution:
(a) The Binomial pmf is $f(x|\theta) = \binom{4}{x}\theta^x(1-\theta)^{4-x}$. The likelihood ratio is

$$\frac{f(x|0.8)}{f(x|0.5)} = \frac{(0.8)^x(0.2)^{4-x}}{(0.5)^4} = \frac{0.8^x \cdot 0.2^{4-x}}{0.0625}.$$

Computing for each value of x:

x	0	1	2	3	4
$f(x\|0.5)$	1/16	4/16	6/16	4/16	1/16
$f(x\|0.8)$	0.0016	0.0256	0.1536	0.4096	0.4096
Ratio	0.0256	0.1024	0.4096	1.6384	6.5536

Equivalently, the ratio is $\frac{(0.8)^x(0.2)^{4-x}}{(0.5)^4} = 16 \cdot 4^x \cdot 0.2^4 = 4^x \cdot 0.0256$. This is increasing in x, confirming that the NP test rejects for large x.
(b) For size $\alpha = 1/16$, we need $P_{0.5}(\text{reject}) = 1/16$. Since $P_{0.5}(X = 4) = (0.5)^4 = 1/16$, the rejection region is $\{X = 4\}$.

(c) The power at $\theta = 0.8$ is

$$P_{0.8}(X = 4) = (0.8)^4 = 0.4096.$$

(d) We want size exactly $\alpha = 5/16$. Note that $P_{0.5}(X \geq 4) = 1/16$ and $P_{0.5}(X \geq 3) = 5/16$. Since $5/16$ is exactly achievable, the (non-randomized) UMP test rejects when $X \geq 3$.
To verify: $P_{0.5}(X \geq 3) = P_{0.5}(X = 3) + P_{0.5}(X = 4) = 4/16 + 1/16 = 5/16 = \alpha$.
If the target size fell strictly between $1/16$ and $5/16$—say $\alpha = 3/16$—then we would need a randomized test: reject with certainty when $X = 4$ (contributing $1/16$), and reject with probability γ when $X = 3$ (contributing $\gamma \cdot 4/16$). Setting $1/16 + 4\gamma/16 = 3/16$ gives $\gamma = 1/2$. But for $\alpha = 5/16$ exactly, no randomization is needed.
The power of the $\alpha = 5/16$ test at $\theta = 0.8$ is

$$P_{0.8}(X \geq 3) = P_{0.8}(X = 3) + P_{0.8}(X = 4) = 4 \cdot (0.8)^3(0.2) + (0.8)^4 = 0.4096 + 0.4096 = 0.8192.$$

Drill: Sample Size Determination

You are testing $H_0 \colon \mu \leq 100$ vs. $H_1 \colon \mu > 100$ with $\sigma = 15$ known.

(a) Find n so that $\alpha = 0.05$ and power at $\mu = 105$ is at least 0.80.

(b) Repeat for power at least 0.90.

(c) Repeat for power at least 0.95 at $\mu = 103$.

Solution:
The sample size formula for a one-sided z-test with known σ is

$$n = \frac{(z_\alpha + z_\beta)^2 \, \sigma^2}{(\mu_1 - \mu_0)^2},$$

where $\mu_0 = 100$, $\sigma = 15$, and $1 - \beta$ is the desired power (so z_β is the upper β quantile of the standard normal).
(a) Here $\alpha = 0.05$, power $= 0.80$ (so $\beta = 0.20$), and $\mu_1 = 105$, giving $\delta = 5$.
We need $z_{0.05} = 1.645$ and $z_{0.20} = 0.842$.

$$n = \frac{(1.645 + 0.842)^2 \cdot 225}{25} = \frac{(2.487)^2 \cdot 225}{25} = \frac{6.185 \cdot 225}{25} = \frac{1391.6}{25} \approx 55.7.$$

Rounding up: $n = 56$.
(b) Now power $= 0.90$, so $\beta = 0.10$ and $z_{0.10} = 1.282$.

$$n = \frac{(1.645 + 1.282)^2 \cdot 225}{25} = \frac{(2.927)^2 \cdot 225}{25} = \frac{8.567 \cdot 225}{25} = \frac{1927.6}{25} \approx 77.1.$$

Rounding up: $n = 78$.
(c) Now power $= 0.95$ (so $\beta = 0.05$, $z_{0.05} = 1.645$) and $\mu_1 = 103$, giving $\delta = 3$.

$$n = \frac{(1.645 + 1.645)^2 \cdot 225}{9} = \frac{(3.290)^2 \cdot 225}{9} = \frac{10.824 \cdot 225}{9} = \frac{2435.4}{9} \approx 270.6.$$

Rounding up: $n = 271$.

Drill: p-Value Computation

In a sample of $n = 20$ from $\mathrm{n}(\mu, 9)$, you observe $\bar{x} = 1.2$.

(a) Compute the p-value for $H_0\colon \mu \le 0$ vs. $H_1\colon \mu > 0$.

(b) Compute the p-value for $H_0\colon \mu = 0$ vs. $H_1\colon \mu \ne 0$.

(c) At which significance levels ($\alpha = 0.01, 0.05, 0.10$) would you reject in each case?

Solution:
We have $n = 20$, $\sigma^2 = 9$ (so $\sigma = 3$), and $\bar{x} = 1.2$. Since σ is known, we use the z-test. The test statistic is

$$z = \frac{\bar{x} - \mu_0}{\sigma/\sqrt{n}} = \frac{1.2 - 0}{3/\sqrt{20}} = \frac{1.2}{3/4.472} = \frac{1.2}{0.6708} \approx 1.789.$$

(a) For the one-sided test $H_0\colon \mu \le 0$ vs. $H_1\colon \mu > 0$, the p-value is the probability of seeing a test statistic at least as extreme under H_0:

$$p = P(Z \ge 1.789) = 1 - \Phi(1.789) \approx 0.0368.$$

(b) For the two-sided test $H_0\colon \mu = 0$ vs. $H_1\colon \mu \ne 0$, the p-value doubles (accounting for both tails):

$$p = 2\,P(Z \ge |1.789|) = 2(1 - \Phi(1.789)) \approx 2 \times 0.0368 = 0.0736.$$

(c) We reject H_0 when $p \le \alpha$.
For the one-sided test ($p \approx 0.037$):

- $\alpha = 0.01$: $0.037 > 0.01$, do not reject.
- $\alpha = 0.05$: $0.037 < 0.05$, reject.
- $\alpha = 0.10$: $0.037 < 0.10$, reject.

For the two-sided test ($p \approx 0.074$):

- $\alpha = 0.01$: $0.074 > 0.01$, do not reject.
- $\alpha = 0.05$: $0.074 > 0.05$, do not reject.
- $\alpha = 0.10$: $0.074 < 0.10$, reject.

7 Interval Estimation — Problems

7.1 Location Family Pivots

Example: Normal CI for μ (σ^2 Known)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$ with σ^2 known.
Step 1: Identify the pivot. The quantity

$$Z = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}}$$

has distribution $\mathrm{n}(0, 1)$, which does not depend on μ. So Z is a pivot.
Step 2: Bound the pivot. For a $1 - \alpha$ confidence interval, choose the symmetric cutoffs $\pm z_{\alpha/2}$:

$$\mathbb{P}\left(-z_{\alpha/2} \leq \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \leq z_{\alpha/2}\right) = 1 - \alpha.$$

Step 3: Invert for μ. Multiply through by $\sigma/\sqrt{n}$ and rearrange:

$$\mathbb{P}\left(\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right) = 1 - \alpha.$$

The $1 - \alpha$ confidence interval for μ is:

$$\left[\bar{X} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}},\ \bar{X} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right].$$

Example: Normal CI for μ (σ^2 Unknown)—The t-Interval

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$ with σ^2 unknown.
When σ^2 is unknown, the Z-pivot no longer works because it contains the unknown σ. The fix is to replace σ with the sample standard deviation $S = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2}$.
Step 1: Identify the pivot. The quantity

$$T = \frac{\bar{X} - \mu}{S/\sqrt{n}}$$

has Student's t distribution with $n - 1$ degrees of freedom. This distribution depends only on n, not on μ or σ^2. So T is a pivot.
Why does this work? Two key facts from normal theory:

1. $\bar{X}$ and S^2 are independent (a consequence of $\bar{X}$ and $X_i - \bar{X}$ being independent in the normal family).
2. $(n - 1)S^2/\sigma^2 \sim \chi^2_{n-1}$.

Therefore

$$T = \frac{\bar{X} - \mu}{\sigma/\sqrt{n}} \cdot \frac{1}{\sqrt{S^2/\sigma^2}} = \frac{Z}{\sqrt{V/(n-1)}},$$

where $Z \sim \mathrm{n}(0,1)$ and $V = (n-1)S^2/\sigma^2 \sim \chi^2_{n-1}$ are independent. This is exactly the definition of a t_{n-1} random variable. The σ^2 cancels.
Step 2: Bound and invert. Using $t_{n-1,\alpha/2}$ cutoffs:

$$\mathbb{P}\left(-t_{n-1,\alpha/2} \le \frac{\bar{X}-\mu}{S/\sqrt{n}} \le t_{n-1,\alpha/2}\right) = 1-\alpha.$$

The $1-\alpha$ confidence interval for μ is:

$$\left[\bar{X} - t_{n-1,\alpha/2}\frac{S}{\sqrt{n}},\ \bar{X} + t_{n-1,\alpha/2}\frac{S}{\sqrt{n}}\right].$$

7.2 Scale Family Pivots

Example: Exponential CI for the Mean λ

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Exponential}(\lambda)$, where $\mathbb{E}[X_i] = \lambda$.
The sufficient statistic is $T = \sum_{i=1}^n X_i$, and $T \sim \text{Gamma}(n, \lambda)$.
Step 1: Identify the pivot. In the gamma pdf, t and λ appear together as t/λ. Define

$$Q(T,\lambda) = \frac{2T}{\lambda}.$$

Since $T \sim \text{Gamma}(n,\lambda)$, we have $2T/\lambda \sim \text{Gamma}(n,2) = \chi^2_{2n}$.
This distribution does not depend on λ, so Q is a pivot.
Step 2: Bound the pivot. Choose χ^2 cutoffs:

$$\mathbb{P}\left(\chi^2_{2n,1-\alpha/2} \le \frac{2T}{\lambda} \le \chi^2_{2n,\alpha/2}\right) = 1-\alpha.$$

Here $\chi^2_{2n,\alpha/2}$ denotes the upper $\alpha/2$ quantile of χ^2_{2n}.
Step 3: Invert for λ. Since $Q = 2T/\lambda$ is decreasing in λ, inverting the inequalities flips the direction:

$$\mathbb{P}\left(\frac{2T}{\chi^2_{2n,\alpha/2}} \le \lambda \le \frac{2T}{\chi^2_{2n,1-\alpha/2}}\right) = 1-\alpha.$$

The $1-\alpha$ confidence interval for λ is:

$$\left[\frac{2\sum X_i}{\chi^2_{2n,\alpha/2}},\ \frac{2\sum X_i}{\chi^2_{2n,1-\alpha/2}}\right].$$

Numerical check. For $n = 10$ and observed $\sum x_i = 47$, a 95% CI uses $\chi^2_{20,0.025} = 34.17$ and $\chi^2_{20,0.975} = 9.59$:

$$\left[\frac{94}{34.17},\ \frac{94}{9.59}\right] = [2.75,\ 9.80].$$

Example: Chi-Squared Pivot for Normal Variance σ^2

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$ with μ unknown.
Step 1: Identify the pivot. The quantity

$$Q = \frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1},$$

where $S^2 = \frac{1}{n-1}\sum_{i=1}^n (X_i - \bar{X})^2$. This is a scale-family pivot: we divided by σ^2.
Step 2: Bound the pivot.

$$\mathbb{P}\left(\chi^2_{n-1,1-\alpha/2} \le \frac{(n-1)S^2}{\sigma^2} \le \chi^2_{n-1,\alpha/2}\right) = 1 - \alpha.$$

Step 3: Invert for σ^2. Since Q is decreasing in σ^2:

$$\mathbb{P}\left(\frac{(n-1)S^2}{\chi^2_{n-1,\alpha/2}} \le \sigma^2 \le \frac{(n-1)S^2}{\chi^2_{n-1,1-\alpha/2}}\right) = 1 - \alpha.$$

Taking square roots gives the CI for σ:

$$\left[\sqrt{\frac{(n-1)S^2}{\chi^2_{n-1,\alpha/2}}}, \sqrt{\frac{(n-1)S^2}{\chi^2_{n-1,1-\alpha/2}}}\right].$$

7.3 CI–Test Duality

Example: Duality for the Normal Mean (Two-Sided)

Consider testing $H_0 : \mu = \mu_0$ vs. $H_1 : \mu \neq \mu_0$ with $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$, σ^2 known.
Test side: The level α test rejects when $|\bar{X} - \mu_0| > z_{\alpha/2} \cdot \sigma/\sqrt{n}$. The acceptance region is

$$A(\mu_0) = \left\{\mathbf{x} : \mu_0 - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \bar{x} \le \mu_0 + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\}.$$

Invert: The confidence set is

$$C(\mathbf{x}) = \left\{\mu_0 : \bar{x} - z_{\alpha/2}\frac{\sigma}{\sqrt{n}} \le \mu_0 \le \bar{x} + z_{\alpha/2}\frac{\sigma}{\sqrt{n}}\right\}.$$

This is exactly the standard Z-interval. The two perspectives are connected by the tautology

$$\mathbf{x} \in A(\mu_0) \iff \mu_0 \in C(\mathbf{x}).$$

Example: One-Sided Test $\Rightarrow$ One-Sided CI

Testing $H_0 : \mu = \mu_0$ vs. $H_1 : \mu < \mu_0$ (reject for small $\bar{X}$):
The test rejects when $\bar{X} < \mu_0 - t_{n-1,\alpha} \cdot S/\sqrt{n}$.
The acceptance region is $A(\mu_0) = \{\mathbf{x} : \bar{x} \ge \mu_0 - t_{n-1,\alpha} \cdot s/\sqrt{n}\}$.
Inverting: $C(\mathbf{x}) = \{\mu_0 : \mu_0 \le \bar{x} + t_{n-1,\alpha} \cdot s/\sqrt{n}\} = (-\infty,\ \bar{x} + t_{n-1,\alpha} \cdot s/\sqrt{n}]$.

This is a one-sided upper confidence bound for μ.

7.4 Two-Sample Pivots

Example: Two-Sample t-Interval (Equal Variances)

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \mathrm{n}(\mu_X, \sigma^2)$ and $Y_1, \ldots, Y_m \overset{\text{iid}}{\sim} \mathrm{n}(\mu_Y, \sigma^2)$, independent samples with common variance σ^2.
Define the pooled sample variance:

$$S_p^2 = \frac{1}{n+m-2}\left(\sum_{i=1}^{n}(X_i - \bar{X})^2 + \sum_{j=1}^{m}(Y_j - \bar{Y})^2\right).$$

The pivot for $\mu_X - \mu_Y$ is:

$$T = \frac{(\bar{X} - \bar{Y}) - (\mu_X - \mu_Y)}{S_p\sqrt{\frac{1}{n} + \frac{1}{m}}} \sim t_{n+m-2}.$$

The $1-\alpha$ CI for $\mu_X - \mu_Y$ is:

$$(\bar{X} - \bar{Y}) \pm t_{n+m-2,\alpha/2} \cdot S_p\sqrt{\frac{1}{n} + \frac{1}{m}}.$$

Example: F-Interval for Variance Ratio

With the same setup as above, the pivot for σ_X^2/σ_Y^2 (when variances are not assumed equal) is:

$$F = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{n-1,m-1}.$$

The $1-\alpha$ CI for σ_X^2/σ_Y^2 is:

$$\left[\frac{S_X^2}{S_Y^2} \cdot \frac{1}{F_{n-1,m-1,\alpha/2}},\ \frac{S_X^2}{S_Y^2} \cdot F_{m-1,n-1,\alpha/2}\right].$$

Note the subscript swap: $F_{m-1,n-1,\alpha/2} = 1/F_{n-1,m-1,1-\alpha/2}$.

7.5 Drill Problems

Drill: Location Pivot—Normal Mean, σ Known

Let $X_1, \ldots, X_{25} \overset{\text{iid}}{\sim} \mathrm{n}(\mu, 9)$. Construct a 95% CI for μ.
Solution. The pivot is $Z = \frac{\bar{X}-\mu}{3/\sqrt{25}} = \frac{\bar{X}-\mu}{0.6} \sim \mathrm{n}(0,1)$.
Using $z_{0.025} = 1.96$:

$$\bar{X} - 1.96(0.6) \le \mu \le \bar{X} + 1.96(0.6).$$

$$\bar{X} \pm 1.176.$$

Drill: Location-Scale Pivot—Normal Mean, σ Unknown

Let $X_1, \ldots, X_{10} \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$, both unknown. Construct a 90% CI for μ.
Solution. The pivot is $T = \frac{\bar{X}-\mu}{S/\sqrt{10}} \sim t_9$.
Using $t_{9,0.05} = 1.833$:

$$\bar{X} - 1.833 \cdot \frac{S}{\sqrt{10}} \leq \mu \leq \bar{X} + 1.833 \cdot \frac{S}{\sqrt{10}}.$$

Drill: Scale Pivot—Normal Variance

Let $X_1, \ldots, X_{16} \overset{\text{iid}}{\sim} \mathrm{n}(\mu, \sigma^2)$. Construct a 95% CI for σ^2.
Solution. The pivot is $Q = \frac{15S^2}{\sigma^2} \sim \chi^2_{15}$.
Using $\chi^2_{15,0.025} = 27.49$ and $\chi^2_{15,0.975} = 6.262$:

$$\frac{15S^2}{27.49} \leq \sigma^2 \leq \frac{15S^2}{6.262}.$$

Drill: Scale Pivot—Exponential

Let $X_1, \ldots, X_8 \overset{\text{iid}}{\sim} \text{Exponential}(\theta)$ where $\mathbb{E}[X_i] = \theta$. Construct a 95% CI for θ.
Solution. The pivot is $Q = \frac{2\sum X_i}{\theta} \sim \chi^2_{16}$.
Using $\chi^2_{16,0.025} = 28.85$ and $\chi^2_{16,0.975} = 6.908$:

$$\frac{2\sum X_i}{28.85} \leq \theta \leq \frac{2\sum X_i}{6.908}.$$

Drill: CI–Test Duality

You have a 95% CI for μ: $[3.2, 5.8]$. Without any further calculation, what is the conclusion of testing $H_0 : \mu = 4$ vs. $H_1 : \mu \neq 4$ at $\alpha = 0.05$? What about $H_0 : \mu = 6$?
Solution. $\mu_0 = 4$ is inside the interval, so fail to reject H_0.
$\mu_0 = 6$ is outside the interval, so reject H_0.
The CI is the set of all μ_0 values for which the two-sided test does not reject.

Drill: Construct a Pivot from Scratch

Let $X_1, \ldots, X_n \overset{\text{iid}}{\sim} \text{Uniform}(0, \theta)$. Find a pivot and construct a $1-\alpha$ CI for θ.
Solution. Let $Y = X_{(n)} = \max(X_1, \ldots, X_n)$. The cdf of Y is $F_Y(y) = (y/\theta)^n$ for $0 \leq y \leq \theta$.
Define the pivot $Q = Y/\theta$. Then Q has pdf $f_Q(q) = nq^{n-1}$ on $[0, 1]$, which does not depend on θ.
Choose a, b such that $\mathbb{P}(a \leq Y/\theta \leq b) = 1 - \alpha$.
Since Q takes values in $[0, 1]$, set $b = 1$ (so $Y \leq \theta$ always) and choose $a = \alpha^{1/n}$ so that $\mathbb{P}\left(Q \geq \alpha^{1/n}\right) = 1 - \alpha$.
Inverting $\alpha^{1/n} \leq Y/\theta \leq 1$:

$$Y \leq \theta \leq \frac{Y}{\alpha^{1/n}}.$$

The $1-\alpha$ CI for θ is $\left[Y,\ Y/\alpha^{1/n}\right]$.

Drill: Two-Sample t-Interval

Independent samples: $X_1, \ldots, X_{12} \overset{\text{iid}}{\sim} \mathrm{n}(\mu_X, \sigma^2)$ and $Y_1, \ldots, Y_8 \overset{\text{iid}}{\sim} \mathrm{n}(\mu_Y, \sigma^2)$ with common unknown σ^2. You observe $\bar{x} = 14.2$, $\bar{y} = 11.7$, and $s_p^2 = 5.3$. Construct a 95% CI for $\mu_X - \mu_Y$.
Solution. Degrees of freedom: $12 + 8 - 2 = 18$. Using $t_{18,0.025} = 2.101$:

$$(\bar{x} - \bar{y}) \pm t_{18,0.025} \cdot s_p \sqrt{\frac{1}{12} + \frac{1}{8}}.$$

Compute the standard error:

$$\sqrt{5.3 \cdot \left(\frac{1}{12} + \frac{1}{8}\right)} = \sqrt{5.3 \cdot 0.2083} = \sqrt{1.104} = 1.051.$$

The interval is:

$$2.5 \pm 2.101 \cdot 1.051 = 2.5 \pm 2.208 = [0.292,\ 4.708].$$

Drill: F-Test for Variance Ratio

With $s_X^2 = 7.4$ ($n = 10$) and $s_Y^2 = 3.1$ ($m = 15$), construct a 90% CI for σ_X^2/σ_Y^2.
Solution. The pivot is $F = \frac{S_X^2/\sigma_X^2}{S_Y^2/\sigma_Y^2} \sim F_{9,14}$.
Using $F_{9,14,0.05} = 2.65$ and $F_{14,9,0.05} = 3.03$:

$$\frac{S_X^2/S_Y^2}{F_{9,14,0.05}} \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{S_X^2}{S_Y^2} \cdot F_{14,9,0.05}.$$

$$\frac{7.4/3.1}{2.65} \le \frac{\sigma_X^2}{\sigma_Y^2} \le \frac{7.4}{3.1} \cdot 3.03.$$

$$0.901 \le \frac{\sigma_X^2}{\sigma_Y^2} \le 7.233.$$

Since the interval contains 1, we cannot reject $H_0 : \sigma_X^2 = \sigma_Y^2$ at the 10% level.

Drill: Pivot Identification

For each scenario, identify whether to subtract, divide, or both:

1. $X_i \sim \mathrm{n}(\mu, 4)$: Pivot is $\bar{X} - \mu$ (subtract—location).
2. $X_i \sim \text{Exponential}(\beta)$: Pivot is $\sum X_i/\beta$ (divide—scale).
3. $X_i \sim \mathrm{n}(\mu, \sigma^2)$, both unknown: Pivot is $(\bar{X} - \mu)/(S/\sqrt{n})$ (subtract and divide—location-scale).
4. $X_i \sim \text{Uniform}(0, \theta)$: Pivot is $X_{(n)}/\theta$ (divide—scale).
5. $X_i \sim \mathrm{n}(0, \sigma^2)$: Pivot is $\sum X_i^2/\sigma^2$ (divide—scale, and $\sum X_i^2/\sigma^2 \sim \chi_n^2$).

8 Linear Regression — Problems

8.1 OLS Derivation

Drill: Derive the Normal Equations

Starting from $\text{RSS} = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$:

(a) Expand the quadratic form and differentiate with respect to $\boldsymbol{\beta}$.

(b) Show that $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$.

(c) Verify that $\mathbf{X}^T\mathbf{e} = \mathbf{0}$ (residuals are orthogonal to all regressors).

Solution:
(a) Expand the RSS quadratic form. Let $\text{RSS}(\boldsymbol{\beta}) = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$. Distributing the product gives

$$\text{RSS}(\boldsymbol{\beta}) = \mathbf{Y}^T\mathbf{Y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{Y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}.$$

Here we used $\mathbf{Y}^T\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{Y}$ since both are scalars.
Differentiating with respect to $\boldsymbol{\beta}$ and setting the gradient to zero:

$$\frac{\partial\,\text{RSS}}{\partial\boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{Y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0}.$$

This yields the **normal equations**:

$$\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{Y}.$$

(b) Since $\mathbf{X}$ is full rank, $\mathbf{X}^T\mathbf{X}$ is invertible. Multiplying both sides of the normal equations by $(\mathbf{X}^T\mathbf{X})^{-1}$ on the left:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}.$$

(c) The residual vector is $\mathbf{e} = \mathbf{Y} - \mathbf{X}\hat{\boldsymbol{\beta}}$. Multiplying on the left by $\mathbf{X}^T$:

$$\mathbf{X}^T\mathbf{e} = \mathbf{X}^T\mathbf{Y} - \mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}}.$$

Substituting the normal equations $\mathbf{X}^T\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}^T\mathbf{Y}$ directly gives

$$\mathbf{X}^T\mathbf{e} = \mathbf{X}^T\mathbf{Y} - \mathbf{X}^T\mathbf{Y} = \mathbf{0}.$$

This says the residual vector is orthogonal to every column of $\mathbf{X}$. In particular, if $\mathbf{X}$ includes an intercept column of ones, then $\mathbf{1}^T\mathbf{e} = 0$, so the residuals sum to zero.

8.2 Distribution Theory

Example: Why Independence of $\hat{\beta}$ and S^2 Matters

The independence of $\hat{\boldsymbol{\beta}}$ and S^2 is what lets us construct t-statistics. Without it, the ratio $(\hat{\beta}_j - \beta_j)/(S\sqrt{d_{jj}})$ would not follow a t-distribution.
The key to the proof: $\hat{\boldsymbol{\beta}}$ depends on $\mathbf{Y}$ through $\mathbf{HY}$, while RSS depends on $\mathbf{Y}$ through $(\mathbf{I}_n - \mathbf{H})\mathbf{Y}$. For normal $\mathbf{Y}$, these are independent because $\mathbf{H}(\mathbf{I}_n - \mathbf{H}) = \mathbf{0}$.

8.3 Hypothesis Testing

Example: Connection: t-Test Is a Special Case of the F-Test

When $q = 1$ (testing a single constraint $H_0 : \beta_j = 0$), the F-statistic equals T^2, where T is the t-statistic. This is because $F_{1,n-p} = t^2_{n-p}$.
Concretely: if $\mathbf{a}^T = (0, \ldots, 0, 1, 0, \ldots, 0)$ selects β_j, then $F = T^2 = \hat{\beta}_j^2/(S^2 d_{jj})$.

Drill: F-Test Computation

Suppose you fit two models:

- **Full model** ($p = 4$ parameters): RSS $= 100$, $n = 30$.
- **Reduced model** ($p_0 = 2$ parameters): $\text{RSS}_H = 150$.

(a) Compute the F-statistic for testing whether the two dropped variables are jointly zero.

(b) State the degrees of freedom.

(c) If $F^{0.05}_{2,26} = 3.37$, do you reject at the 5% level?

Answer: $F = \frac{(150-100)/2}{100/26} = \frac{25}{3.846} = 6.50$. Degrees of freedom: $q = 2$ and $n - p = 26$. Since $6.50 > 3.37$, reject H_0.

8.4 Confidence and Prediction Intervals

Example: Simple Linear Regression Intervals

For simple linear regression $Y = \beta_0 + \beta_1 x + \varepsilon$ with n observations, we have:

$$v_0 = \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{\sum_{i=1}^n (x_i - \bar{x})^2}.$$

Both the CI and PI are **narrowest at** $x_0 = \bar{x}$ (where $v_0 = 1/n$) and widen as x_0 moves away from $\bar{x}$. This is why extrapolation is dangerous: predictions far from the center of the data have very wide intervals.

Drill: CI vs. PI Width

Suppose you have a simple linear regression with $n = 50$, $S = 3$, $\bar{x} = 10$, $\sum(x_i - \bar{x})^2 = 200$, and you want to make an inference at $x_0 = 12$.

(a) Compute v_0.

(b) Write the 95% CI for $\mathbb{E}[Y|x_0 = 12]$.

(c) Write the 95% PI for a new Y_0 at $x_0 = 12$.

(d) Which is wider, and by approximately what factor?

Answer: $v_0 = 1/50 + 4/200 = 0.02 + 0.02 = 0.04$. Using $t^{0.025}_{48} \approx 2.01$: CI half-width $= 2.01 \times 3\sqrt{0.04} = 2.01 \times 0.6 = 1.21$. PI half-width $= 2.01 \times 3\sqrt{1.04} = 2.01 \times 3.06 = 6.15$. The

PI is about 5 times wider.

8.5 Key Interview Topics

Drill: Hat Matrix Properties

(a) Show $\mathbf{H}$ is symmetric and idempotent.

(b) Show $\mathrm{tr}(\mathbf{H}) = p$.

(c) Show that $0 \le h_{ii} \le 1$.

(d) If $n = 20$ and $p = 4$, what is the average leverage? At what threshold would you flag a point as high-leverage?

Solution:
(a) Recall the hat matrix $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$.
Symmetry. Taking the transpose:

$$\mathbf{H}^T = \left[\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\right]^T = \mathbf{X}\left[(\mathbf{X}^T\mathbf{X})^{-1}\right]^T\mathbf{X}^T.$$

Since $\mathbf{X}^T\mathbf{X}$ is symmetric, so is its inverse. Therefore $\mathbf{H}^T = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = \mathbf{H}$.
Idempotency. Compute $\mathbf{H}^2$:

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T \cdot \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T.$$

The middle terms collapse: $\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} = \mathbf{I}_p$. This gives

$$\mathbf{H}^2 = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T = \mathbf{H}.$$

(b) For any idempotent matrix, the trace equals the rank. Since $\mathrm{tr}(\mathbf{AB}) = \mathrm{tr}(\mathbf{BA})$, we can cycle the factors:

$$\mathrm{tr}(\mathbf{H}) = \mathrm{tr}\big(\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\big) = \mathrm{tr}\big((\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\big) = \mathrm{tr}(\mathbf{I}_p) = p.$$

(c) Since $\mathbf{H}$ is idempotent, its eigenvalues are all 0 or 1. Every diagonal entry h_{ii} of $\mathbf{H}$ can be written as

$$h_{ii} = \mathbf{e}_i^T\mathbf{H}\mathbf{e}_i,$$

where $\mathbf{e}_i$ is the i-th standard basis vector. Because $\mathbf{H}$ is positive semidefinite (all eigenvalues are 0 or 1), this gives $h_{ii} \ge 0$.
For the upper bound, idempotency means $\mathbf{H}^2 = \mathbf{H}$, so element-wise:

$$h_{ii} = \sum_{j=1}^{n} h_{ij}^2.$$

Since h_{ii}^2 is one of the terms in the sum, we have $h_{ii} \ge h_{ii}^2$, which means $h_{ii}(1 - h_{ii}) \ge 0$. Therefore $0 \le h_{ii} \le 1$.
(d) Average leverage $= p/n = 4/20 = 0.2$; flag at $> 2(p/n) = 0.4$.

8.6 Comprehensive Drills

Drill: Matrix OLS Derivation

Given the model $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ with $\mathbf{X}$ full rank:

(a) Derive $\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}$ by minimizing RSS.

(b) Show $\mathbb{E}\left[\hat{\boldsymbol{\beta}}\right] = \boldsymbol{\beta}$.

(c) Show $\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}$.

(d) Show $\mathbb{E}[S^2] = \sigma^2$ where $S^2 = \mathrm{RSS}/(n-p)$.

Solution:
(a) We minimize $\mathrm{RSS} = (\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})^T(\mathbf{Y} - \mathbf{X}\boldsymbol{\beta})$. Expanding:

$$\mathrm{RSS} = \mathbf{Y}^T\mathbf{Y} - 2\boldsymbol{\beta}^T\mathbf{X}^T\mathbf{Y} + \boldsymbol{\beta}^T\mathbf{X}^T\mathbf{X}\boldsymbol{\beta}.$$

Setting the gradient to zero:

$$\frac{\partial\,\mathrm{RSS}}{\partial\boldsymbol{\beta}} = -2\mathbf{X}^T\mathbf{Y} + 2\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \mathbf{0}.$$

Since $\mathbf{X}$ is full rank, $\mathbf{X}^T\mathbf{X}$ is invertible, giving

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{Y}.$$

(b) Substitute $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ into the OLS formula:

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T(\mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}).$$

Distributing gives

$$\hat{\boldsymbol{\beta}} = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}.$$

Taking expectations, and using $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$:

$$\mathbb{E}\left[\hat{\boldsymbol{\beta}}\right] = \boldsymbol{\beta} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbb{E}[\boldsymbol{\varepsilon}] = \boldsymbol{\beta}.$$

So $\hat{\boldsymbol{\beta}}$ is unbiased.
(c) From part (b), $\hat{\boldsymbol{\beta}} - \boldsymbol{\beta} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\boldsymbol{\varepsilon}$. Let $\mathbf{A} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$ for convenience. Then

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \mathbf{A}\ \mathrm{Var}(\boldsymbol{\varepsilon})\ \mathbf{A}^T.$$

Using $\mathrm{Var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_n$:

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2\mathbf{A}\mathbf{A}^T = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}.$$

The middle terms collapse to $\mathbf{I}_p$, leaving

$$\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1}.$$

(d) Write RSS $= \mathbf{e}^T\mathbf{e}$ where $\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\mathbf{Y}$ and $\mathbf{H} = \mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T$. Substituting $\mathbf{Y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$ and using $(\mathbf{I}_n - \mathbf{H})\mathbf{X} = \mathbf{0}$:

$$\mathbf{e} = (\mathbf{I}_n - \mathbf{H})\boldsymbol{\varepsilon}.$$

Let $\mathbf{M} = \mathbf{I}_n - \mathbf{H}$. Note that $\mathbf{M}$ is symmetric and idempotent. Then

$$\text{RSS} = \boldsymbol{\varepsilon}^T\mathbf{M}\boldsymbol{\varepsilon}.$$

Taking expectations using the identity $\mathbb{E}\left[\boldsymbol{\varepsilon}^T\mathbf{A}\boldsymbol{\varepsilon}\right] = \operatorname{tr}(\mathbf{A}\ \operatorname{Var}(\boldsymbol{\varepsilon}))$ (valid when $\mathbb{E}[\boldsymbol{\varepsilon}] = \mathbf{0}$):

$$\mathbb{E}[\text{RSS}] = \operatorname{tr}(\mathbf{M} \cdot \sigma^2\mathbf{I}_n) = \sigma^2 \operatorname{tr}(\mathbf{M}).$$

Since $\operatorname{tr}(\mathbf{M}) = \operatorname{tr}(\mathbf{I}_n) - \operatorname{tr}(\mathbf{H}) = n - p$:

$$\mathbb{E}[\text{RSS}] = \sigma^2(n - p).$$

Dividing both sides by $n - p$ gives $\mathbb{E}[S^2] = \sigma^2$, confirming that S^2 is an unbiased estimator of σ^2.

Drill: Gauss–Markov Proof

Let $\mathbf{a}^T\tilde{\boldsymbol{\beta}} = \mathbf{a}^T\mathbf{C}\mathbf{Y}$ be any linear unbiased estimator of $\mathbf{a}^T\boldsymbol{\beta}$.

(a) Write $\mathbf{C} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \mathbf{D}$. What does unbiasedness require of $\mathbf{D}$?

(b) Show that $\operatorname{Var}\left(\mathbf{a}^T\tilde{\boldsymbol{\beta}}\right) \geq \operatorname{Var}\left(\mathbf{a}^T\hat{\boldsymbol{\beta}}\right)$ for all $\mathbf{a}$.

Solution:
(a) Write $\mathbf{C} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \mathbf{D}$, so that $\tilde{\boldsymbol{\beta}} = \mathbf{C}\mathbf{Y}$. For $\tilde{\boldsymbol{\beta}}$ to be unbiased for $\boldsymbol{\beta}$, we need $\mathbb{E}\left[\tilde{\boldsymbol{\beta}}\right] = \boldsymbol{\beta}$ for every $\boldsymbol{\beta}$. Since $\mathbb{E}[\mathbf{Y}] = \mathbf{X}\boldsymbol{\beta}$:

$$\mathbb{E}\left[\tilde{\boldsymbol{\beta}}\right] = \mathbf{C}\mathbf{X}\boldsymbol{\beta} = \left[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \mathbf{D}\right]\mathbf{X}\boldsymbol{\beta}.$$

The first term gives $(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{X}\boldsymbol{\beta} = \boldsymbol{\beta}$. So unbiasedness requires

$$\mathbf{D}\mathbf{X}\boldsymbol{\beta} = \mathbf{0} \quad \text{for all } \boldsymbol{\beta},$$

which means $\mathbf{D}\mathbf{X} = \mathbf{0}$.
(b) Compute the variance of the alternative estimator. Since $\operatorname{Var}(\mathbf{Y}) = \sigma^2\mathbf{I}_n$:

$$\operatorname{Var}\left(\tilde{\boldsymbol{\beta}}\right) = \sigma^2\mathbf{C}\mathbf{C}^T.$$

Expanding $\mathbf{C}\mathbf{C}^T$ using $\mathbf{C} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \mathbf{D}$:

$$\mathbf{C}\mathbf{C}^T = \left[(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T + \mathbf{D}\right]\left[\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} + \mathbf{D}^T\right].$$

Multiplying out the four terms:

$$\mathbf{C}\mathbf{C}^T = (\mathbf{X}^T\mathbf{X})^{-1} + (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{D}^T + \mathbf{D}\mathbf{X}(\mathbf{X}^T\mathbf{X})^{-1} + \mathbf{D}\mathbf{D}^T.$$

From part (a), $\mathbf{DX} = \mathbf{0}$. Transposing gives $\mathbf{X}^T\mathbf{D}^T = \mathbf{0}$ as well. The two cross terms vanish, leaving

$$\mathrm{Var}\left(\tilde{\boldsymbol{\beta}}\right) = \sigma^2(\mathbf{X}^T\mathbf{X})^{-1} + \sigma^2\mathbf{D}\mathbf{D}^T.$$

The first term is $\mathrm{Var}\left(\hat{\boldsymbol{\beta}}\right)$. The second term $\sigma^2\mathbf{D}\mathbf{D}^T$ is positive semidefinite.
For any vector $\mathbf{a}$:

$$\mathrm{Var}\left(\mathbf{a}^T\tilde{\boldsymbol{\beta}}\right) = \mathbf{a}^T\,\mathrm{Var}\left(\tilde{\boldsymbol{\beta}}\right)\,\mathbf{a} = \mathrm{Var}\left(\mathbf{a}^T\hat{\boldsymbol{\beta}}\right) + \sigma^2\mathbf{a}^T\mathbf{D}\mathbf{D}^T\mathbf{a}.$$

Since $\mathbf{a}^T\mathbf{D}\mathbf{D}^T\mathbf{a} = \|\mathbf{D}^T\mathbf{a}\|^2 \geq 0$:

$$\mathrm{Var}\left(\mathbf{a}^T\tilde{\boldsymbol{\beta}}\right) \geq \mathrm{Var}\left(\mathbf{a}^T\hat{\boldsymbol{\beta}}\right).$$

This proves the Gauss–Markov theorem: OLS has minimum variance among all linear unbiased estimators. Equality holds only when $\mathbf{D}^T\mathbf{a} = \mathbf{0}$, i.e., when the alternative estimator coincides with OLS for the linear combination of interest.

Drill: Confidence vs. Prediction Interval

Explain, in both formulas and intuition:

(a) Why the prediction interval is always wider than the confidence interval.

(b) What happens to each interval as $n \to \infty$?

(c) Why both intervals are widest far from $\bar{x}$ in simple regression.

Solution:
(a) For a new observation at predictor value $\mathbf{x}_0$, define $v_0 = \mathbf{x}_0^T(\mathbf{X}^T\mathbf{X})^{-1}\mathbf{x}_0$. The confidence interval for the mean response $\mathbb{E}[Y_0] = \mathbf{x}_0^T\boldsymbol{\beta}$ has variance

$$\mathrm{Var}\left(\mathbf{x}_0^T\hat{\boldsymbol{\beta}}\right) = \sigma^2 v_0.$$

The prediction interval targets a *new* observation $Y_0 = \mathbf{x}_0^T\boldsymbol{\beta} + \varepsilon_0$. The prediction error is $\hat{Y}_0 - Y_0 = \mathbf{x}_0^T\hat{\boldsymbol{\beta}} - Y_0$, and since ε_0 is independent of $\hat{\boldsymbol{\beta}}$:

$$\mathrm{Var}\left(\hat{Y}_0 - Y_0\right) = \sigma^2 v_0 + \sigma^2 = \sigma^2(1 + v_0).$$

The prediction interval always carries the extra σ^2 term from the irreducible noise in the new observation. No matter how well we estimate the regression line, a single new Y_0 still scatters around it with variance σ^2.
(b) As $n \to \infty$, the estimation uncertainty vanishes: $v_0 \to 0$ because $(\mathbf{X}^T\mathbf{X})^{-1}$ shrinks toward zero.
The confidence interval width shrinks to zero, since its variance $\sigma^2 v_0 \to 0$. With enough data, we pin down the mean response exactly.
The prediction interval width converges to $\pm\, z_{\alpha/2}\,\sigma$ (using the normal quantile in the limit). It does *not* shrink to zero because the irreducible noise σ^2 remains. Even with infinite data, we cannot predict a single future observation perfectly.
(c) In simple regression, $v_0 = \frac{1}{n} + \frac{(x_0-\bar{x})^2}{\sum(x_i-\bar{x})^2}$. The second term grows quadratically as x_0 moves away from $\bar{x}$.

Intuitively, the regression line is "pinned" near $(\bar{x}, \bar{Y})$ and free to rotate at the extremes. Points far from $\bar{x}$ contribute the most leverage, so the fitted line is most uncertain there. Both the CI and PI inherit this extra uncertainty through the v_0 factor, making both intervals widest far from the center of the data.

Drill: R^2 Under the Null

Suppose $\beta_1 = \cdots = \beta_{p-1} = 0$ (all slopes are zero).

(a) Show that $\mathbb{E}[R^2] = (p-1)/(n-1)$. *Hint:* Under H_0, the regression sum of squares $\text{RSS}_H - \text{RSS}$ divided by σ^2 is χ^2_{p-1}, and $\text{TSS}/\sigma^2 \approx \chi^2_{n-1}$.

(b) For $n = 50$ and $p = 6$, what is $\mathbb{E}[R^2]$ under the null?

(c) Why does this make the adjusted R^2 a better model comparison tool?

Solution:
(a) Under H_0 (all slopes zero), define $\text{SSR} = \text{TSS} - \text{RSS}$ (the regression sum of squares). By standard results, $\text{SSR}/\sigma^2 \sim \chi^2_{p-1}$ and $\text{RSS}/\sigma^2 \sim \chi^2_{n-p}$, and the two are independent. Also, $\text{TSS}/\sigma^2 \sim \chi^2_{n-1}$.
Since $R^2 = \text{SSR}/\text{TSS}$, write

$$R^2 = \frac{\text{SSR}/\sigma^2}{\text{TSS}/\sigma^2} = \frac{U}{U+V},$$

where $U = \text{SSR}/\sigma^2 \sim \chi^2_{p-1}$ and $V = \text{RSS}/\sigma^2 \sim \chi^2_{n-p}$ are independent.
By definition, $U/(p-1)$ divided by $V/(n-p)$ follows an $F_{p-1,\,n-p}$ distribution. The quantity $R^2/(1-R^2)$ can be written as U/V, so R^2 is a monotone transformation of this F-statistic.
To find $\mathbb{E}[R^2]$ directly, note that $R^2 = U/(U+V)$ follows a $\text{Beta}(\frac{p-1}{2}, \frac{n-p}{2})$ distribution (since U and V are independent chi-squareds). The mean of this Beta distribution is

$$\mathbb{E}\left[R^2\right] = \frac{(p-1)/2}{(p-1)/2 + (n-p)/2} = \frac{p-1}{n-1}.$$

(b) For $n = 50$ and $p = 6$: $\mathbb{E}[R^2] = 5/49 \approx 0.102$.
(c) Even when no true relationship exists (H_0 holds), R^2 is positive on average — roughly $(p-1)/(n-1)$. Each additional regressor inflates R^2 by about $1/(n-1)$ in expectation, purely from fitting noise. This means raw R^2 can look impressive simply because p is large relative to n.
The adjusted R^2 corrects for this:

$$R^2_{\text{adj}} = 1 - \frac{n-1}{n-p}(1-R^2) = 1 - \frac{\text{RSS}/(n-p)}{\text{TSS}/(n-1)}.$$

Under H_0, $\mathbb{E}[\text{RSS}/(n-p)] = \sigma^2$ and $\text{TSS}/(n-1) \approx \sigma^2$, so $\mathbb{E}\left[R^2_{\text{adj}}\right] \approx 0$. Unlike raw R^2, the adjusted version does not systematically inflate when useless regressors are added. It penalizes model complexity by scaling the residual sum of squares by $1/(n-p)$ rather than $1/n$, making it a more honest tool for comparing models of different sizes.

Drill: Partial F-Test

You run a regression of salary on experience, education, and age ($p = 4$, $n = 100$).

- Full model: $R^2 = 0.45$, RSS $= 550$.
- Model without education and age ($p_0 = 2$): $\text{RSS}_H = 700$.

(a) Compute the partial F-statistic.

(b) State the null hypothesis, numerator df, and denominator df.

(c) Is the result significant at the 5% level? (Use $F_{2,96}^{0.05} \approx 3.09$.)

Answer: $F = \frac{(700-550)/2}{550/96} = \frac{75}{5.729} = 13.09$. H_0: $\beta_{\text{education}} = \beta_{\text{age}} = 0$; df: $(2, 96)$. Since $13.09 > 3.09$, reject at 5%.